NEET UG 2027: Master Chapter Theory

Organic Chemistry: Electronic Effects, Reaction Mechanisms & Biomolecules

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1 NMC Syllabus Demarcation & Zero-Tolerance Deletions

Curriculum Alignment & Pedagogical Scope

The revised National Medical Commission (NMC) NEET UG 2027 Chemistry syllabus demands a rigorous, mechanistic understanding of organic transformations rather than superficial rote memorization. The curriculum is built upon three integrated pillars:

1. **General Organic Chemistry (GOC) & Reaction Intermediates:** Quantitative and qualitative evaluation of electron displacement effects (inductive, electromeric, resonance, hyperconjugation, and aromaticity) and the structural thermodynamics of transient intermediates (carbocations, carbanions, free radicals, carbenes, and arynes).
2. **Systematic Functional Group Reaction Pathways:** Exhaustive mechanism sequences spanning aliphatic/aromatic hydrocarbons, haloalkanes/haloarenes, oxygenated functions (alcohols, phenols, ethers, aldehydes, ketones, carboxylic acids), and nitrogenous compounds (amines, diazonium salts).
3. **Biomolecules of Life:** Structural stereochemistry of carbohydrates, amino acids, peptide linkages, primary through quaternary protein architecture, nucleic acid Watson-Crick geometry, vitamins, and hormones.

Zero-Tolerance NMC Deleted Topics Alert: Do Not Study!

To optimize preparation efficiency and adhere strictly to the regulatory mandate issued by the NMC for NEET UG 2024–2027, the following entire domains are **permanently excluded** from the examination:

- **Polymers (Entire Chapter Deleted):** Classification of polymers; addition and condensation polymerization; natural and synthetic rubber, vulcanization; polythene (LDPE/HDPE), Teflon, PAN, Nylon-6, Nylon-6,6, Terylene/Dacron, Bakelite, melamine-formaldehyde, Buna-S, Buna-N; biodegradable polymers (PHBV, Nylon-2-Nylon-6).
- **Chemistry in Everyday Life (Entire Chapter Deleted):** Drugs and pharmacodynamics; analgesics (narcotic and non-narcotic); tranquilizers, antiseptics (Dettol, Bithionol), disinfectants, antibiotics, antacids, antihistamines; artificial sweeteners (saccharin, aspartame, alitame, sucralose); food preservatives; soaps and synthetic detergents (anionic, cationic, non-ionic).

Candidates must **not** allocate study time to these deleted units. Any occurrence of these keywords in this text serves solely as an exclusion boundary warning.

2 General Organic Chemistry (GOC) & Electronic Effects

2.1 Inductive Effect ($\pm I$)

Physical Origin & Characteristics of Inductive Effect

The inductive effect is a permanent polarization of a σ-bond arising from electronegativity differences between bonded atoms. The more electronegative atom pulls the bonding σ-electron pair toward itself, establishing a static dipole:

$$-\mathrm{C}^{\delta\delta\delta+} - \mathrm{C}^{\delta\delta+} - \mathrm{C}^{\delta+} - \mathrm{X}^{\delta-} \quad (\text{where } \chi_X > \chi_C) \tag{1}$$

Key Physical Attributes:

- **Permanent Distortion:** It involves shifting of σ-electrons without complete bond cleavage or free electron transfer.

- **Distance Attenuation:** The magnitude of the fractional charge decreases rapidly with distance and becomes virtually negligible beyond the third σ-carbon atom (C_3).
- **Hydrogen Reference Standard:** The electronegativity of hydrogen is taken as the arbitrary zero benchmark: $I_H = 0$.

Authoritative $-I$ and $+I$ Relative Strength Series

1. Electron-Withdrawing ($-I$) Series (Decreasing Order of Pull):

$$\begin{aligned} -NF_3^+ > -NR_3^+ > -NH_3^+ > -NO_2 > -SO_3H > -CN > -COOH > -F > -Cl \\ > -Br > -I > -OR > -OH > -C \equiv CR > -NH_2 > -C_6H_5 > -CH = CH_2 > -H \end{aligned} \tag{2}$$

2. Electron-Releasing ($+I$) Series (Decreasing Order of Push):

$$-O^- > -COO^- > -C(CH_3)_3 > -CH(CH_3)_2 > -CH_2CH_3 > -CH_3 > -D > -H \tag{3}$$

Physical Rationales for Key Sequences:

- **Positive Formal Charges:** Positively charged ammonium and nitro centers exhibit dramatically higher $-I$ than neutral halogens because full ionic charges create immense electrostatic attraction.
- **Hybridization Dependency:** Higher $\%s$-character pulls electrons closer to the carbon nucleus, raising electronegativity:

$$sp \text{ carbon } (50\%s, \chi \approx 3.29) > sp^2 \text{ carbon } (33.3\%s, \chi \approx 2.75) > sp^3 \text{ carbon } (25\%s, \chi \approx 2.48) \tag{4}$$

  Hence, alkynyl ($-C \equiv CR$) exerts a stronger $-I$ effect than vinyl ($-CH = CH_2$) and phenyl ($-C_6H_5$).
- **Isotope Effect:** Tritium ($-T$) and Deuterium ($-D$) have shorter and stronger $C-D$ bonds than $C-H$ due to lower zero-point vibrational energy, causing less average nuclear distance and greater electron release: $+I(-T) > +I(-D) > +I(-H)$.

Quantitative Applications: Acidic Strength of Carboxylic Acids

The acidic strength (K_a, where $pK_a = -\log K_a$) of carboxylic acids is determined by the stability of the conjugate carboxylate anion ($R-COO^-$):

$$\text{Acidic Strength} \propto \text{Stability of Conjugate Base } (R-COO^-) \propto \frac{-I \text{ effect}}{+I \text{ effect}} \tag{5}$$

1. Influence of Halogen Identity (Electronegativity):

$$\begin{aligned} FCH_2COOH\ (pK_a = 2.66) > ClCH_2COOH\ (2.86) > BrCH_2COOH\ (2.90) \\ > ICH_2COOH\ (3.17) > CH_3COOH\ (4.76) \end{aligned} \tag{6}$$

2. Number of Electron-Withdrawing Groups:

$$Cl_3C-COOH\ (pK_a = 0.65) > Cl_2CH-COOH\ (1.29) > ClCH_2-COOH\ (2.86) > CH_3COOH\ (4.76) \tag{7}$$

3. Distance Dependence (α, β, γ-Positions):

$$\begin{aligned} CH_3-CH_2-CH(Cl)-COOH\ (\alpha\text{-isomer}, pK_a = 2.84) > CH_3-CH(Cl)-CH_2-COOH\ (\beta, 4.06) \\ > CH_2(Cl)-CH_2-CH_2-COOH\ (\gamma, 4.52) \end{aligned} \tag{8}$$

Direct NCERT Line: Formic Acid vs Benzoic Acid vs Acetic Acid

The electron-donating $+I$ effect of the methyl group destabilizes the acetate anion relative to the formate anion:

$$HCOOH\ (pK_a = 3.75) > C_6H_5COOH\ (pK_a = 4.20) > CH_3COOH\ (pK_a = 4.76) \tag{9}$$

Crucial Insight: Even though phenyl is electron-withdrawing by $-I$ relative to sp^3 alkyl, its $+M$ resonance into the carbonyl group weakens acidity compared to formic acid. Thus, formic acid is stronger than benzoic acid!

2.2 Resonance / Mesomeric Effect ($\pm M$)

The Quantum Mechanical Basis of Resonance

Resonance describes the delocalization of π-electrons or non-bonding lone pairs across parallel, unhybridized p-orbitals in planar conjugated frameworks. **Mandatory Conditions for Conjugation:**

1. **Planarity:** All participating atomic nuclei must lie in a single plane (or near-planar geometry) to permit effective sideways overlap of adjacent p-orbitals.
2. **Conjugated Topologies:** Conjugation occurs across alternating systems:
   - $\pi - \sigma - \pi$ (e.g., 1,3-butadiene, benzene)
   - $\pi - \sigma - p_{\text{vacant}}$ (e.g., allyl carbocation, tropylium cation)
   - $\pi - \sigma -$ lone pair / non-bonding (e.g., vinyl chloride, phenoxide, aniline)
   - $\pi - \sigma -$ unpaired electron ($\bullet$) (e.g., allyl free radical)
   - Lone pair $-\sigma - p_{\text{vacant}}$ (e.g., $CH_3 - \overset{+}{C}H - \ddot{O}CH_3$)

Hierarchical Rules for Evaluating Canonical Structure Stability

When ranking the thermodynamic contribution of contributing resonance structures (canonical forms) to the true resonance hybrid, apply the following hierarchy strictly in order:

1. **Complete Octet Rule (Paramount Criterion):** Structures in which all second-period atoms (C, N, O, F) possess complete valence octets contribute overwhelmingly more than structures with incomplete octets, **even if** a formal positive charge resides on an electronegative atom (O or N):

$$CH_3 - \overset{+}{O} = CH_2 \text{ (all octets complete, Major)} \longleftrightarrow CH_3 - \ddot{O} - \overset{+}{C}H_2 \text{ (carbon has } 6e^-\text{, Minor)} \tag{10}$$

2. **Maximum Number of Covalent Bonds:** Greater number of covalent bonds maximizes attractive nuclear-electronic interactions, conferring greater stability.
3. **Charge Separation vs Uncharged Species:** Neutral structures without formal charge separation are intrinsically more stable than dipolar (zwitterionic) structures.
4. **Electronegativity & Charge Concordance:** Among charge-separated structures:
   - Negative formal charges reside most stably on the most electronegative atoms ($O^- > N^- > C^-$).
   - Positive formal charges reside most stably on the least electronegative atoms ($C^+ > N^+ > O^+$).
5. **Proximity of Like/Unlike Charges:** Canonical forms with like charges on adjacent atoms are violently destabilized (electrostatic repulsion); unlike charges close together confer electrostatic stabilization.

Resonance Energy (E_{res}) Formulation

Resonance energy is defined as the difference in energy between the actual resonance hybrid and the most stable theoretical canonical Lewis structure:

$$E_{\text{res}} = E_{\text{canonical (most stable)}} - E_{\text{actual hybrid}} \tag{11}$$

Experimentally, resonance energy is determined from standard enthalpies of hydrogenation ($\Delta_{\text{hydro}}H^\circ$):

$$\begin{aligned} \text{Cyclohexene} + H_2 &\rightarrow \text{Cyclohexane} \quad \Delta H^\circ = -119.7 \text{ kJ/mol} \\ \text{Theoretical 1,3,5-Cyclohexatriene (no resonance)} &\implies 3 \times (-119.7) = -359.1 \text{ kJ/mol} \\ \text{Experimental Benzene} + 3H_2 &\rightarrow \text{Cyclohexane} \quad \Delta H^\circ = -208.5 \text{ kJ/mol} \\ E_{\text{res}}(\text{Benzene}) &= |-359.1 - (-208.5)| = 150.6 \text{ kJ/mol} \ (\approx 36.0 \text{ kcal/mol}) \end{aligned} \tag{12}$$

NEET Trap: Halogen Paradox in EAS ($+M$ vs $-I$)

Halogens ($-$F, $-$Cl, $-$Br, $-$I) attached to benzene present a classical dual nature:

- **Reactivity (Net Deactivating):** Halogens are strongly electronegative ($\chi_F = 4.0, \chi_{Cl} = 3.16$), creating a powerful $-I$ effect that drains σ-electron density from the ring. The $-I$ effect dominates over the $+M$ resonance effect for overall rate, making haloarenes **less reactive** than benzene toward electrophiles.

- **Regioselectivity (Ortho/Para Directing):** During attack by an incoming electrophile (E^+), the halogen donates its p-lone pair via $+M$ resonance to stabilize the positively charged arenium ion specifically when E^+ attacks the **ortho** or **para** positions (giving a complete-octet canonical structure [$Cl^+ = C_6H_5E$]). At the meta position, the halogen lone pair cannot stabilize the carbocation.

Conclusion: Halogens are uniquely **deactivating ortho/para-directors** in EAS.

2.3 Hyperconjugation (Baker-Nathan Effect)

The $\sigma \to \pi^*$ and $\sigma \to p$ No-Bond Resonance

Hyperconjugation involves the delocalization of bonding electrons in a C – H (or C – C) σ-bond into an adjacent vacant or partially filled non-bonding p-orbital or an antibonding π^* molecular orbital:

$$\sigma_{C-H} \longrightarrow p_{vacant} \quad \text{(in carbocations)} \quad \text{or} \quad \sigma_{C-H} \longrightarrow \pi^* \quad \text{(in alkenes)} \tag{13}$$

Structural Characteristics:

1. **Number of Contributing Structures:** For an alkene, carbocation, or free radical with n α-hydrogens (hydrogens on the sp^3 carbon directly attached to the sp^2 system), the total number of hyperconjugative canonical forms is:

$$\text{Total Hyperconjugative Structures} = n + 1 \quad (\text{where } n = \text{number of } \alpha\text{-H}) \tag{14}$$

2. **Bond Shortening & Bond Lengthening:** The single bond between the sp^3 and sp^2 carbons acquires partial double bond character (e.g., the C – C single bond in propene is 1.49 Å, significantly shorter than standard 1.54 Å in ethane). The C_α–H bond is lengthened and weakened.
3. **Reverse Hyperconjugation:** Observed when electronegative halogen atoms reside at the α-carbon (e.g., $-CX_3$ in trifluoromethylbenzene $C_6H_5 - CF_3$). Electron density flows from the π-system into the vacant σ^*_{C-F} antibonding orbital, causing meta-direction and ring deactivation.

Alkene Stability & Heat of Hydrogenation ($\Delta_{hydro}H^\circ$)

Alkene thermodynamic stability is directly proportional to the number of hyperconjugative α-hydrogens:

$$\text{Alkene Stability} \propto (\text{Number of } \alpha\text{-H}) \propto \frac{1}{\text{Heat of Hydrogenation } (\Delta_{hydro}H^\circ)} \tag{15}$$

Authoritative Stability Sequence:

$$\begin{aligned} (CH_3)_2C=C(CH_3)_2\ (12\alpha\text{-H}) &> (CH_3)_2C=CHCH_3\ (9\alpha\text{-H}) \\ &> CH_2=C(CH_3)_2\ (6\alpha\text{-H, gem-dimethyl}) \\ &> \textit{trans}\text{–}CH_3CH=CHCH_3\ (6\alpha\text{-H}) \\ &> \textit{cis}\text{–}CH_3CH=CHCH_3\ (6\alpha\text{-H}) \\ &> CH_3CH_2CH=CH_2\ (2\alpha\text{-H}) \\ &> CH_2=CH_2\ (0\alpha\text{-H}) \end{aligned} \tag{16}$$

The *trans* vs *cis* Isomer Trap: *trans*-2-butene is more stable than *cis*-2-butene because the two bulky methyl groups are oriented at 180°, eliminating the steric van der Waals repulsion present in the eclipsed-like geometry of the *cis*-isomer. The *gem*-dimethyl alkene ($CH_2=C(CH_3)_2$) is thermodynamically more stable than both *cis*- and *trans*-2-butene due to superior electrostatic distribution across the two methyl groups attached to the same sp^2 carbon.

2.4 Electromeric Effect ($\pm E$)

Temporary Reagent-Induced π-Polarization

The electromeric effect is a purely **temporary** electronic displacement operating exclusively in molecules containing multiple bonds ($C=C, C\equiv C, C=O$) upon the approach of an attacking polar reagent. Upon removal of the reagent, the system instantly reverts to its unpolarized ground state.

- **Positive Electromeric Effect ($+E$):** The π-electron pair shifts toward the atom to which the attacking reagent attaches:

$$CH_2=CH_2 + H^+ \xrightarrow{+E} CH_2^+ - CH_2 - H \quad \text{(Electrophilic attack on alkene)} \tag{17}$$

- **Negative Electromeric Effect ($-E$):** The π-electron pair shifts away from the atom to which the attacking reagent attaches:

$$>C=O + CN^- \xrightarrow{-E} >C(CN) - O^- \quad \text{(Nucleophilic attack on carbonyl)} \tag{18}$$

2.5 Aromaticity & Hückel's Rule

The Four Cardinal Criteria for Aromaticity

To be classified as **aromatic**, a chemical entity must simultaneously satisfy all four structural and electronic requirements formulated by Erich Hückel:

1. **Cyclic Structure:** The system must possess a continuous closed loop of bonded atoms.
2. **Planar Geometry:** All ring vertices must be strictly planar (or possess < 0.1 Å deviation) to maintain parallel p-orbital orientation.
3. **Complete Unbroken Conjugation:** Every atom in the cyclic ring must possess an unhybridized p-orbital (i.e., fully sp^2 or sp hybridized; zero tetrahedral sp^3 carbons in the ring loop).
4. **Hückel's $(4n+2)\pi$ Electron Rule:** The number of delocalized π-electrons must equal $(4n+2)$, where n is a non-negative integer ($n = 0, 1, 2, 3, \cdots \implies 2, 6, 10, 14, 18\ldots$ π-electrons).

Aromatic vs Anti-Aromatic vs Non-Aromatic Classification

Property	Aromatic	Anti-Aromatic	Non-Aromatic
Cyclic & Planar?	Yes	Yes	Lacks cyclicity or planarity
Conjugation	Unbroken throughout ring	Unbroken throughout ring	Interrupted by sp^3 carbon
π-Electron Count	$(4n+2)\pi$ electrons	$4n\pi$ electrons ($n = 1, 2, \ldots$)	Arbitrary count
Thermodynamic Stability	Exceptionally High ($E_{res} \gg 0$)	Exceptionally Low ($E_{strain} \gg 0$)	Normal open-chain analog
Magnetic Induced Field	Diatropic (Diamagnetic ring current)	Paratropic (Paramagnetic ring current)	No ring current
Canonical Examples	Benzene (6π), Tropylium (6π)	Cyclobutadiene (4π), Cyclopentadienyl cation (4π)	1,3-Cyclohexadiene, COT (tub-shaped)

Annulenes & Non-Planar Ring Escapes

1. Cyclooctatetraene (COT, [8]-Annulene): COT possesses 8π-electrons ($4n$ with $n=2$). If planar, it would be severely anti-aromatic. To escape extreme anti-aromatic electronic destabilization, the flexible 8-membered ring distorts into a non-planar **tub conformation**. Because p-orbitals are perpendicular at the edges, conjugation is broken: COT is strictly **non-aromatic**. However, reduction with potassium metal yields the cyclooctatetraenyl dianion ($C_8H_8^{2-}$), which has 10π-electrons ($4n+2, n=2$), becomes strictly planar, and is **aromatic**!

2. [10]-Annulene: Possesses 10π-electrons ($4n+2, n=2$). In the planar configuration, severe steric repulsion between the two internal hydrogen atoms forces the ring out of planarity $\implies$ **non-aromatic**. If the internal hydrogens are replaced with a methylene bridge (1,6-methano[10]annulene), the ring achieves planarity and exhibits robust **aromaticity**.

Heterocyclic Aromaticity & Lone Pair Involvement

When determining Hückel π-electron counts in heterocycles containing N, O, S:

- **Pyridine (C_5H_5N):** The nitrogen atom is sp^2-hybridized. Its unhybridized p-orbital contributes $1e^-$ to the 6π aromatic sextet. The lone pair resides in an sp^2 hybrid orbital in the ring plane, perpendicular to the π-cloud. It does **not** participate in aromaticity $\implies$ Pyridine is basic ($pK_b \approx 8.8$).
- **Pyrrole (C_4H_4NH):** The nitrogen atom is sp^2-hybridized. Its lone pair resides in an unhybridized p-orbital, contributing $2e^-$ to complete the 6π aromatic sextet. The lone pair is tied up in aromaticity $\implies$ Pyrrole is an exceptionally weak base ($pK_b \approx 13.6$) and protonation on nitrogen destroys aromaticity.
- **Furan (C_4H_4O) & Thiophene (C_4H_4S):** The heteroatom contributes one lone pair ($2e^-$) to the aromatic sextet (6π electrons); the second lone pair resides in an sp^2 orbital orthogonal to the π-system.

3 Reactive Intermediates & Molecular Rearrangements

3.1 Carbocations

Geometry, Electronic Configuration & Stability Hierarchy

Carbocations are trivalent carbon cations possessing a positive formal charge and a sextet ($6e^-$) of valence electrons:

- **Geometry & Hybridization:** The positively charged carbon is sp^2-hybridized with trigonal planar geometry (120° bond angles). The remaining unhybridized $2p_z$ orbital is completely vacant.
- **Stability Factors:** Stabilized by $+M$ resonance delocalization, aromatic stabilization, hyperconjugation ($n_{\alpha-\mathrm{H}}$), and $+I$ inductive effects.

$$\begin{aligned}\text{Tropylium } (6\pi\text{-aromatic}) &\approx \text{Cyclopropylcarbinyl (dancing resonance)}\\ &> (C_6H_5)_3C^+ > (C_6H_5)_2CH^+ > C_6H_5CH_2^+ \approx CH_2=CH-CH_2^+\\ &> 3^\circ\ (CH_3)_3C^+\ (9\alpha\text{-H}) > 2^\circ\ (CH_3)_2CH^+\ (6\alpha\text{-H})\\ &> 1^\circ\ CH_3CH_2^+\ (3\alpha\text{-H}) > CH_3^+\ (0\alpha\text{-H}) > \text{vinyl } (CH_2=CH^+) > \text{phenyl } (C_6H_5^+)\end{aligned} \tag{19}$$

The Driving Force & Pathways of Carbocation Rearrangements

A carbocation will spontaneously rearrange if and only if the rearranged carbocation possesses lower Gibbs free energy (greater thermodynamic stability). Rearrangements occur via intramolecular 1,2-migrations of a hydride, alkyl, or aryl group into the adjacent vacant p-orbital:

1. **1,2-Hydride Shift (H^-):**

$$CH_3-CH_2-CH_2^+\ (1^\circ, 2\alpha\text{-H}) \xrightarrow{1,2\text{-}H^-} CH_3-\overset{+}{C}H-CH_3\ (2^\circ, 6\alpha\text{-H}) \tag{20}$$

2. **1,2-Methyl / Alkyl Shift (R^-):**

$$(CH_3)_3C-\overset{+}{C}H_2\ (\text{neopentyl}, 1^\circ) \xrightarrow{1,2\text{-}CH_3^-} CH_3-\overset{+}{C}(CH_3)-CH_2CH_3\ (3^\circ, 8\alpha\text{-H}) \tag{21}$$

3. **Ring Expansion (Angle Strain Relief):** Four- and five-membered rings with an exocyclic carbocation undergo instantaneous 1,2-bond migration to expand the ring:
   - **Cyclobutylcarbinyl Cation → Cyclopentyl Cation:** The 90° ring angle expands to 108°, relieving ≈ 108.8 kJ/mol of Baeyer angle strain.
   - **Cyclopentylcarbinyl Cation → Cyclohexyl Cation:** Expansion from 5-membered ring (108°) to strain-free chair cyclohexane (109.5°).

NEET Trap: Migratory Aptitude in Carbocation Rearrangements

When competing groups are attached to the migrating carbon, the intrinsic migratory aptitude is:

$$\mathrm{H}^- > \text{Aryl } (\mathrm{Ar}^-) > \text{Alkyl } (3^\circ > 2^\circ > 1^\circ > \mathrm{CH}_3^-) \tag{22}$$

For substituted aryl groups: electron-donating groups accelerate migration:

$$\begin{aligned} p\text{-anisyl } (p\text{–}\mathrm{CH_3O} - \mathrm{C_6H_4}\text{–}) > p\text{-tolyl } (p\text{–}\mathrm{CH_3} - \mathrm{C_6H_4}\text{–}) > \text{phenyl } (\mathrm{C_6H_5}\text{–}) \\ > p\text{-chlorophenyl} > p\text{-nitrophenyl} \end{aligned} \tag{23}$$

Crucial Rule: Never perform a less-stable to less-stable rearrangement unless it immediately opens a direct pathway to an overwhelmingly more stable intermediate (e.g., resonance-delocalized benzylic or allylic system).

3.2 Carbanions

Structure, Pyramidal Inversion & Stability Hierarchy

Carbanions possess a trivalent carbon bearing a formal negative charge and eight valence electrons (3 bonding pairs + 1 lone pair):

- **Hybridization & Inversion:** Unconjugated carbanions are sp^3-hybridized with trigonal pyramidal geometry. The lone pair undergoes rapid umbrella inversion (flipping barrier $\approx$ 25 kJ/mol), making isolated chiral carbanions rapidly racemize.
- **Conjugated Carbanions (sp^2 Planarity):** If the carbanion is adjacent to a π-system or an electron-withdrawing carbonyl group ($-\mathrm{C=O}$), it rehybridizes to sp^2 to accommodate the lone pair in an unhybridized p-orbital for maximum resonance delocalization.
- **Stability Sequence:**

$$\begin{aligned} \text{Cyclopentadienyl Anion } (6\pi) > (\mathrm{C_6H_5})_3\mathrm{C}^- > (\mathrm{C_6H_5})_2\mathrm{CH}^- \\ > \mathrm{C_6H_5CH_2^-} \approx \mathrm{CH_2=CH-CH_2^-} > \mathrm{HC\equiv C^-} \ (sp, 50\%s) \\ > \mathrm{CH_2=CH^-} \ (sp^2) \approx \mathrm{C_6H_5^-} > \mathrm{CH_3^-} > 1^\circ > 2^\circ > 3^\circ \end{aligned} \tag{24}$$

Active Methylene Compounds & Enolate Chemistry

Methylene groups ($-\mathrm{CH_2}-$) flanked between two strong electron-withdrawing groups (e.g., diethyl malonate, ethyl acetoacetate, acetylacetone) possess exceptionally acidic hydrogens ($pK_a \approx 9-13$):

$$\mathrm{CH_3-CO-CH_2-CO-CH_3 + EtO^- \rightleftharpoons [CH_3-CO-\bar{C}H-CO-CH_3 \longleftrightarrow 2\ Enolate\ Forms] + EtOH} \tag{25}$$

The resulting enolate anion is an **ambident nucleophile** that can attack electrophiles via carbon (*C*-alkylation) or oxygen (*O*-alkylation), driving fundamental $\mathrm{C-C}$ bond-forming reactions (Aldol, Claisen, Michael additions).

3.3 Free Radicals

Structure, Generation & Allylic Halogenation via NBS

Carbon free radicals are neutral trivalent species possessing seven valence electrons (one unpaired electron):

- **Structure:** Planar sp^2 hybridized with the odd electron occupying the unhybridized $2p$ orbital, or shallow rapidly inverting pyramid.
- **Stability:** Stabilized by hyperconjugation and resonance delocalization:

$$(\mathrm{C_6H_5})_3\mathrm{C}^\bullet > (\mathrm{C_6H_5})_2\dot{\mathrm{C}}\mathrm{H} > \mathrm{C_6H_5}\dot{\mathrm{C}}\mathrm{H_2} \approx \mathrm{CH_2=CH-}\dot{\mathrm{C}}\mathrm{H_2} > 3^\circ > 2^\circ > 1^\circ > \dot{\mathrm{C}}\mathrm{H_3} > \text{vinyl} \tag{26}$$

- **N-Bromosuccinimide (NBS) Allylic/Benzylic Halogenation:** NBS selectively brominates the al-

lylic or benzylic position without adding across the C = C double bond. **Mechanism:** Trace HBr reacts with NBS to maintain a constant, extremely low concentration of molecular bromine (Br_2):

$$Br^{\bullet} + CH_3 - CH = CH_2 \longrightarrow HBr + [\dot{C}H_2 - CH = CH_2 \longleftrightarrow CH_2 = CH - \dot{C}H_2] \quad \text{(RDS)} \tag{27}$$

$$NBS + HBr \longrightarrow \text{Succinimide} + Br_2 \quad (\text{Maintains } [Br_2] \approx 10^{-6} \text{ M}) \tag{28}$$

$$\dot{C}H_2 - CH = CH_2 + Br_2 \longrightarrow BrCH_2 - CH = CH_2 + Br^{\bullet} \tag{29}$$

The low $[Br_2]$ prevents electrophilic addition across the double bond, ensuring exclusive allylic radical substitution!

3.4 Carbenes, Nitrenes & Arynes

Singlet vs Triplet Carbenes (: CR_2)

Feature	Singlet Carbene (^{1} : CR_2)	Triplet Carbene (^{3} : CR_2)
Electron Spin State	Spin-paired ($\uparrow\downarrow$, $S = 0, 2S + 1 = 1$)	Parallel spins ($\uparrow\uparrow$, $S = 1, 2S + 1 = 3$)
Hybridization & Geometry	sp^2, Bent ($102 - 108^\circ$)	sp linear (180°) or bent sp^2 ($130 - 150^\circ$)
Orbital Configuration	Lone pair in sp^2, vacant p-orbital	One electron in sp^2, one in p-orbital
Relative Stability	Thermodynamically less stable (except halocarbenes : CCl_2)	Ground state for : CH_2 (Hund's rule)
Addition to Alkenes	**Stereospecific syn-addition** (concerted, geometry retained: cis → cis)	**Non-stereospecific** (diradical intermediate, spin flip allows rotation → cis + trans)

Generation of Dichlorocarbene & The Benzyne Mechanism

1. Dichlorocarbene (: CCl_2) via α-Elimination: Chloroform treated with strong alkali undergoes two-step α-elimination:

$$CHCl_3 + OH^- \overset{\text{fast}}{\rightleftharpoons} : \bar{C}Cl_3 + H_2O \xrightarrow[\text{RDS}]{\text{loss of } Cl^-} : CCl_2 \quad \text{(Electrophilic singlet carbene)} \tag{30}$$

: CCl_2 serves as the key electrophilic intermediate in the **Reimer-Tiemann reaction** and the **Carbylamine reaction**.

2. Benzyne Intermediate (Elimination-Addition): Aryl halides lacking electron-withdrawing groups react with extremely strong bases (e.g., KNH_2/liq.NH_3) via elimination of HX to yield **benzyne** (C_6H_4):

- **Bonding in Benzyne:** The formal triple bond is formed by sideways overlap of two sp^2 hybrid orbitals in the plane of the ring. This bond does not disturb the 6π aromatic sextet, but overlap is extremely weak due to poor orbital alignment $\implies$ severe ring strain and immense reactivity.
- **Cine-Substitution:** Addition of NH_3 across the strained triple bond occurs with equal probability at either carbon, producing isotopic scrambling (50% direct substitution + 50% cine-substitution).

4 Hydrocarbons: Aliphatic & Aromatic Pathways

4.1 Alkanes

Free-Radical Halogenation Kinetics & Hammond's Postulate

Free radical halogenation follows a three-stage chain mechanism (Initiation, Propagation, Termination):

$$\text{Initiation: } X_2 \xrightarrow{h\nu \text{ or } \Delta} 2X^\bullet \tag{31}$$

$$\text{Propagation: } R-H + X^\bullet \xrightarrow{k_1} R^\bullet + HX \quad \text{(RDS for bromination)} \tag{32}$$

$$R^\bullet + X_2 \xrightarrow{k_2} R-X + X^\bullet \tag{33}$$

Relative Reactivity Rates at 298 K:

- **Chlorination (Reactivity over Selectivity):** $1^\circ : 2^\circ : 3^\circ = 1.0 : 3.8 : 5.0$. Chlorination is highly exothermic; the transition state is early (reactant-like per Hammond's postulate), meaning $C-H$ bond cleavage has barely progressed, yielding low regioselectivity.
- **Bromination (Selectivity over Reactivity):** $1^\circ : 2^\circ : 3^\circ = 1.0 : 82 : 1600$. Bromination is endothermic; the transition state is late (product-like), with extensive radical character on carbon, leading to massive selectivity for the 3° alkyl halide ($> 99\%$).

Conformational Analysis of Ethane and n-Butane

- **Ethane:** Infinite conformations between **staggered** (dihedral angle $\theta = 60^\circ$, minimum energy) and **eclipsed** ($\theta = 0^\circ$, maximum energy). The energy barrier is 12.5 kJ/mol (3.0 kcal/mol), caused by **torsional strain** (electron cloud repulsion between adjacent $C-H$ bonds).
- **n-Butane (Rotation about $C_2 - C_3$):**

$$\text{Anti } (\theta = 180^\circ) > \text{Gauche } (\theta = 60^\circ) > \text{Partially Eclipsed } (\theta = 120^\circ) > \text{Fully Eclipsed } (\theta = 0^\circ) \tag{34}$$

  The anti conformation is the global minimum. Gauche is destabilized by 3.8 kJ/mol due to van der Waals steric repulsion between the two bulky methyl groups. The fully eclipsed form is destabilized by 19.0 kJ/mol due to combined torsional and steric strain.
- **Gauche Exception (Intramolecular H-Bonding):** In ethylene glycol ($HO-CH_2-CH_2-OH$) and 2-fluoroethanol ($F-CH_2-CH_2-OH$), the **gauche** conformation is more stable than anti due to stabilizing intramolecular hydrogen bonding.

4.2 Alkenes

Electrophilic Addition of Halogens: Cyclic Halonium Ion

Electrophilic addition of Br_2 (in nonpolar solvent CCl_4) to alkenes proceeds via a cyclic three-membered **bromonium ion**:

1. The π-cloud of the alkene polarizes the $Br-Br$ bond, displacing the distal bromide ion to form a bridged cyclic bromonium ion intermediate.
2. The nucleophilic Br^- attacks the cyclic intermediate from the **backside** (anti-attack) with complete inversion at the attacked carbon.
3. **Stereospecificity:**
   - *trans*-Alkene + Br_2 (anti-addition) $\longrightarrow$ ***meso*-dibromide**.
   - *cis*-Alkene + Br_2 (anti-addition) $\longrightarrow$ **racemic mixture ($\pm$) of enantiomers**.
   - *(Mnemonic: CAR – Cis + Anti → Racemic; TAM – Trans + Anti → Meso).*

Markovnikov Addition vs Anti-Markovnikov Peroxide Effect

1. Markovnikov's Rule: In electrophilic addition of unsymmetrical HX to unsymmetrical alkenes, the electrophilic proton (H^+) attaches to the sp^2 carbon carrying the greater number of hydrogen atoms to generate the more stable carbocation intermediate (which may rearrange!).
2. Anti-Markovnikov (Kharasch / Peroxide Effect): Operates exclusively when **HBr** is added in the

presence of organic peroxides ($R-O-O-R$):

$$\text{Initiation: } ROOR \xrightarrow{\Delta} 2RO^{\bullet}; \quad RO^{\bullet} + HBr \rightarrow ROH + Br^{\bullet} \tag{35}$$

$$\text{Propagation 1: } CH_3-CH=CH_2 + Br^{\bullet} \rightarrow CH_3-\dot{C}H-CH_2Br \quad (\Delta H^{\circ} = -42 \text{ kJ/mol}) \tag{36}$$

$$\text{Propagation 2: } CH_3-\dot{C}H-CH_2Br + HBr \rightarrow CH_3-CH_2-CH_2Br + Br^{\bullet} \quad (\Delta H^{\circ} = -38 \text{ kJ/mol}) \tag{37}$$

Thermochemical Reason: Why Peroxide Effect Operates ONLY for HBr

NEET frequently tests why the peroxide effect fails completely for HCl and HI:

- **Failure with HCl:** The $H-Cl$ bond is exceptionally strong (431 kJ/mol). Step 2 ($R^{\bullet} + HCl \rightarrow R-H + Cl^{\bullet}$) is strongly **endothermic** ($\Delta H^{\circ} \approx +27$ kJ/mol). The chain reaction cannot propagate.
- **Failure with HI:** The $H-I$ bond is weak (297 kJ/mol), making $I^{\bullet}$ generation easy. However, iodine radicals are large and unreactive toward alkenes: Step 1 (Alkene $+ I^{\bullet} \rightarrow$ Radical) is **endothermic** ($\Delta H^{\circ} \approx +54$ kJ/mol). Instead of adding to the double bond, iodine radicals dimerize instantaneously: $2I^{\bullet} \rightarrow I_2$.

Conclusion: Only for HBr are **both propagation steps exothermic**, permitting chain reaction!

Comparative Summary: The Three Alkene Hydration Pathways

Parameter	Acid-Catalyzed Hydration	Hydroboration-Oxidation (HBO)	Oxymercuration-Demercuration (OMDM)
Reagents	H_2O/H_2SO_4	1. $BH_3 \cdot THF$, 2. H_2O_2, OH^-	1. $Hg(OAc)_2, H_2O$, 2. $NaBH_4$
Regiochemistry	Markovnikov	**Anti-Markovnikov**	**Markovnikov**
Stereochemistry	Non-stereospecific	**Syn-addition**	Overall non-stereospecific (anti in step 1)
Intermediate	Free Carbocation	4-center cyclic transition state	Cyclic mercurinium ion
Rearrangement?	**Yes (frequent)**	**Zero rearrangement**	**Zero rearrangement**
Propene Product	Propan-2-ol	Propan-1-ol	Propan-2-ol
3,3-Dimethylbut-1-ene	2,3-dimethylbutan-2-ol (rearranged)	3,3-dimethylbutan-1-ol	3,3-dimethylbutan-2-ol (clean Markovnikov)

Direct NCERT Extraction: Ozonolysis & Structure Elucidation

Ozonolysis cleaves the $C=C$ double bond into two carbonyl fragments:

- **Reductive Ozonolysis (O_3 followed by Zn/H_2O or $(CH_3)_2S$):** Aldehydes and ketones are preserved. Zinc destroys hydrogen peroxide ($H_2O_2 + Zn \rightarrow ZnO + H_2O$), preventing oxidation of aldehydes to carboxylic acids.
- **Oxidative Ozonolysis (O_3 followed by H_2O_2 or H_2O):** All aldehydes are oxidized to carboxylic acids ($-CHO \rightarrow -COOH$); terminal $=CH_2$ becomes $CO_2 + H_2O$; ketones remain unchanged.

4.3 Alkynes

Terminal Alkyne Acidity & Selective Hydrogenation

1. Acidity of Terminal Alkynes ($-C \equiv CH$): The *sp*-hybridized carbon has 50%*s*-character, placing valence electrons closest to the carbon nucleus. Terminal alkynes are acidic ($pK_a \approx 25$):

- React with sodamide: $R-C \equiv CH + NaNH_2 \rightarrow R-C \equiv C^-Na^+ + NH_3 \uparrow$.
- Tollens' Test: Forms a **white precipitate** of silver acetylide: $R-C \equiv CH + [Ag(NH_3)_2]^+OH^- \rightarrow R-C \equiv C-Ag \downarrow + 2NH_3 + H_2O$.

- Ammoniacal Cu_2Cl_2: Forms a **red precipitate** of cuprous acetylide: $R-C\equiv CH + [Cu(NH_3)_2]^+OH^- \rightarrow R-C\equiv C-Cu\downarrow + 2NH_3 + H_2O$.
- *Note: Internal alkynes ($R-C\equiv C-R'$) give negative tests with both reagents!*

2. Stereoselective Reduction to Alkenes:

- **Lindlar's Catalyst ($H_2/Pd-CaCO_3$ poisoned with quinoline / lead acetate):** Syn-addition of hydrogen yielding ***cis*-alkene**.
- **Birch Reduction (Na/liq. NH_3 at $-33°C$):** Anti-addition proceeding via radical-anion intermediate with trans-geometry to minimize electron-electron repulsion, yielding ***trans*-alkene**.

3. Kucherov Hydration ($HgSO_4$/dil.H_2SO_4 at 333 K): Electrophilic addition of water across the triple bond yields an enol, which immediately undergoes keto-enol tautomerism:

$$HC\equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} [CH_2=CH-OH] \rightleftharpoons CH_3-CHO \quad \text{(Acetaldehyde, only aldehyde formed)}$$
$$CH_3-C\equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} [CH_3-C(OH)=CH_2] \rightleftharpoons CH_3-CO-CH_3 \quad \text{(Acetone)} \tag{38}$$

4.4 Aromatic Hydrocarbons: EAS Mechanics & Substituent Effects

Fundamental 3-Step EAS Mechanism & Arenium Ion (σ-Complex)

Electrophilic aromatic substitution (S_EAr) preserves the massive resonance stabilization of the benzene ring (150.6 kJ/mol):

1. **Generation of Active Electrophile (E^+):** Lewis or Brønsted acid generates E^+ ($NO_2^+, Cl^+, SO_3, R^+, RCO^+$).
2. **Formation of Arenium Ion (σ-Complex / Wheland Intermediate) [RDS]:** $C_6H_6 + E^+ \xrightarrow{\text{slow}} [C_6H_6E]^+$. The attacked carbon rehybridizes from sp^2 to sp^3, disrupting aromaticity. Positive charge delocalizes over ortho and para positions.
3. **Rapid Deprotonation & Rearomatization:** Fast abstraction of the proton from the sp^3 carbon restores the planar sp^2 aromatic sextet.

Directing and Activating Effects of Ring Substituents

Group Type	Examples	Electronic Mechanism	Directing Influence
Strongly Activating	$-OH$, $-O^-$, $-NH_2$, $-NR_2$	Dominant $+M$ resonance	Ortho / Para
Moderately Activating	$-OCH_3$, $-NHCOCH_3$, $-OCOR$	$+M$ modulated by cross-conjugation	Ortho / Para
Weakly Activating	$-CH_3$, $-CH_2CH_3$, $-C_6H_5$	Hyperconjugation & $+I$	Ortho / Para
Deactivating (Halogens)	$-F$, $-Cl$, $-Br$, $-I$	Strong $-I$ dominates rate; $+M$ directs	**Ortho / Para**
Moderately Deactivating	$-CHO$, $-COR$, $-COOH$, $-COOR$	$-M$ resonance & $-I$	Meta
Strongly Deactivating	$-NO_2$, $-CF_3$, $-CCl_3$, $-CN$, $-SO_3H$, $-NR_3^+$	Powerful $-M$ / $-I$ withdraws electron density	Meta

Friedel-Crafts Reactions: Critical Limitations & Exceptions

1. Carbocation Rearrangement in Alkylation: Reaction of benzene with 1-chloropropane ($CH_3CH_2CH_2Cl$) and $AlCl_3$ yields **isopropylbenzene (cumene)** as major product (65 – 70%) due to 1,2-hydride shift of the primary propyl carbocation to the more stable secondary carbocation!

$$\text{Benzene} + CH_3CH_2CH_2Cl \xrightarrow{AlCl_3} C_6H_5-CH(CH_3)_2 \text{ (Major)} + C_6H_5-CH_2CH_2CH_3 \text{ (Minor)} \tag{39}$$

Solution: To synthesize linear alkylbenzenes, carry out **Friedel-Crafts Acylation** with propanoyl chloride to yield ethyl phenyl ketone (no rearrangement because acylium ion $R-C\equiv O^+$ is resonance-stabilized),

followed by **Clemmensen reduction** ($Zn - Hg/HCl$) or **Wolff-Kishner reduction**.
2. Polyalkylation vs Monoacylation: Alkyl groups activate the ring toward further substitution (polyalkylation occurs). Acyl groups ($-COR$) are strongly deactivating, ensuring clean monoacylation.
3. Complete Failure with Aniline and Phenol: Aniline ($-NH_2$) and Phenol ($-OH$) fail Friedel-Crafts reactions because their lone pairs form an insoluble Lewis acid-base adduct with $AlCl_3$ (e.g., $Ph - NH_2 \cdot AlCl_3$), placing a positive charge adjacent to the ring and rendering it deactivated toward electrophiles!

5 Haloalkanes & Haloarenes

5.1 Nucleophilic Aliphatic Substitution (S_N1 vs S_N2 vs S_Ni)

Comprehensive Mechanistic Comparison: S_N1 vs S_N2

Parameter	S_N1 (Substitution Nucleophilic Unimolecular)	S_N2 (Substitution Nucleophilic Bimolecular)
Number of Steps	Two discrete elementary steps	Single concerted step
Rate Law & Kinetics	Rate $= k[R - X]$ (First order)	Rate $= k[R - X][Nu^-]$ (Second order)
Substrate Reactivity	$3^\circ > 2^\circ \gg 1^\circ > CH_3$ (governed by carbocation stability)	$CH_3 > 1^\circ > 2^\circ \gg 3^\circ$ (governed by steric hindrance)
Transition State	Involves carbocation intermediate	Pentacoordinate trigonal bipyramidal TS
Stereochemical Outcome	Partial racemization with net inversion ($50 - 70\%$ inversion due to front-side ion pair shielding)	**100% Complete Walden Inversion** (backside attack)
Nucleophile Strength	Weak, neutral nucleophiles (H_2O, ROH) suffice	Requires strong, anionic nucleophiles (OH^-, RO^-, CN^-)
Solvent Preference	**Polar Protic** (H_2O, EtOH, AcOH): solvates carbocation and stabilizes leaving group anion	**Polar Aprotic** (DMSO, DMF, acetone): solvates cations, leaving naked, highly reactive anions
Rearrangements	Carbocation rearrangements (shifts, expansions) common	**Zero rearrangement possible**
Leaving Group Order	$I^- > Br^- > Cl^- \gg F^-$	$I^- > Br^- > Cl^- \gg F^-$

The S_Ni Mechanism: Darzens Process with Thionyl Chloride

Reaction of alcohols with $SOCl_2$ proceeds with different stereochemical outcomes depending on the solvent:

- **In Dioxane / Diethyl Ether (S_Ni, Internal Return):** Reaction forms an alkyl chlorosulfite ester ($R - O - SOCl$). In nonpolar solvent, it dissociates into an intimate ion pair [R^+ SO_2Cl^-]. Decomposition releases $SO_2 \uparrow$, and the chloride is delivered to the carbocation from the **same face** from which the oxygen departed. Outcome: **Retention of configuration**.
- **In Pyridine (S_N2 Inversion):** Pyridine reacts with generated HCl and converts the chlorosulfite into an active pyridinium intermediate. Free chloride ion attacks from the **backside**. Outcome: **Complete Walden inversion**.

5.2 Elimination Reactions ($E1$ vs $E2$ vs $E1cB$)

Saytzeff (Zaitsev) vs Hofmann Regioselectivity

1. Saytzeff Rule (Thermodynamic Control): Dehydrohalogenation yields preferentially the more stable, more highly substituted alkene (greater number of hyperconjugative α-hydrogens).

- **Conditions for Saytzeff Selectivity:** Small, unhindered bases (EtO^-, MeO^-, OH^-) acting on sub-

strates with good leaving groups (I^-, Br^-, Cl^-).

$$CH_3 - CH_2 - CH(Br) - CH_3 \xrightarrow{\text{alc. KOH}} CH_3 - CH = CH - CH_3 \text{ (Saytzeff, 81\%)} + CH_3 - CH_2 - CH = CH_2 \text{ (Hofmann, 19\%)} \quad (40)$$

2. Hofmann Rule (Kinetic Control): Elimination yields preferentially the less substituted, terminal alkene.

- **Conditions for Hofmann Selectivity:**
  1. **Bulky Sterically Hindered Bases:** Potassium tert-butoxide ($t-BuO^-K^+$), diisopropylamide. The base cannot access the more hindered internal β-hydrogen and abstracts the accessible terminal β-hydrogen.
  2. **Poor Leaving Groups:** Fluoride ($-F$) or quaternary ammonium ($-NR_3^+$) and sulfonium ($-SR_2^+$) salts. Due to extreme electronegativity, the β-hydrogen becomes acidic, developing significant carbanion character in the transition state. The less substituted primary carbanion is more stable than secondary, directing elimination to the terminal alkene.

3. $E2$ **Stereochemistry:** Elimination requires strict **anti-periplanar** geometry: the $\beta-H$, the two carbons, and the leaving group X must lie in a single plane with dihedral angle $\theta = 180°$.

5.3 Nucleophilic Aromatic Substitution (S_NAr)

Addition-Elimination via Meisenheimer Complex

Unactivated haloarenes are virtually inert toward nucleophiles at ambient conditions because the $C-X$ bond possesses partial double bond character (1.69 Å in chlorobenzene vs 1.77 Å in chloroethane) due to $+M$ resonance, and the sp^2 carbon holds electrons tightly. **Activation by Nitro Groups (S_NAr):** Introducing strong electron-withdrawing nitro ($-NO_2$) groups at **ortho** and **para** positions dramatically enhances substitution rate:

$$\text{Chlorobenzene (623 K, 300 atm)} \gg p\text{-nitro (443 K)} \gg 2,4\text{-dinitro (368 K)} \gg 2,4,6\text{-trinitro (warm } H_2O) \quad (41)$$

Mechanism: Nucleophilic attack on the ipso carbon forms a resonance-stabilized carbanionic intermediate (**Meisenheimer complex**), where negative charge delocalizes into the ortho and para nitro groups. Loss of halide in Step 2 restores aromaticity. **Leaving Group Anomaly:** Unlike aliphatic substitutions, the rate order in S_NAr is:

$$Ar - F \gg Ar - Cl > Ar - Br > Ar - I \quad (42)$$

The rate-determining step is nucleophilic attack (Step 1). The extreme electronegativity of fluorine stabilizes the Meisenheimer intermediate via $-I$ pull, accelerating the rate!

6 Alcohols, Phenols & Ethers

6.1 Alcohols: Acidity, Oxidation & Dehydration

Acidity & The Lucas Distinction Test

1. Acidity Order of Alcohols:

$$H_2O\ (pK_a = 15.7) > CH_3OH\ (15.5) > 1°\ RCH_2OH\ (16) > 2°\ R_2CHOH\ (16.5) > 3°\ R_3COH\ (17) \quad (43)$$

Alkoxide anions ($R-O^-$) are destabilized by the $+I$ inductive effect of alkyl groups. Water is a stronger acid than all alcohols except methanol!

2. The Lucas Test (anhydrous $ZnCl_2$ + conc. HCl): Distinguishes $1°, 2°, 3°$ alcohols via S_N1 conversion to insoluble alkyl chlorides:

$$R - OH + HCl \xrightarrow{ZnCl_2} R - Cl \downarrow \text{ (cloudy turbidity)} + H_2O \quad (44)$$

- $3°$ **Alcohols:** Immediate turbidity within 5–10 seconds at room temperature.

- 2° **Alcohols:** Turbidity appears within 5 minutes.
- 1° **Alcohols:** Solution remains completely clear at room temperature; turbidity appears only on prolonged boiling.

Oxidation Reagents for Alcohols

- **Pyridinium Chlorochromate (PCC in CH_2Cl_2):** Mild selective oxidant. Oxidizes 1° alcohols to **aldehydes** (stops cleanly without over-oxidation to carboxylic acid) and 2° alcohols to ketones.
- **Jones Reagent (CrO_3/aq. H_2SO_4/acetone):** Converts 1° alcohols directly to carboxylic acids; 2° alcohols to ketones.
- **Heated Copper Catalyst at 573 K (300°C):**
  - 1° Alcohol $\xrightarrow{\text{Cu,573K}}$ Aldehyde $+H_2\uparrow$ (Dehydrogenation).
  - 2° Alcohol $\xrightarrow{\text{Cu,573K}}$ Ketone $+H_2\uparrow$ (Dehydrogenation).
  - 3° Alcohol $\xrightarrow{\text{Cu,573K}}$ **Alkene** $+H_2O$ (**Dehydration**, not dehydrogenation!).

6.2 Phenols: Acidic Character & Named Reactions

Phenol Acidity & Substituent Influence

Phenol ($pK_a \approx 10$) is 10^6 times more acidic than aliphatic alcohols ($pK_a \approx 16-18$):

- **Resonance Stabilization of Phenoxide:** The phenoxide ion ($C_6H_5O^-$) delocalizes negative charge into the benzene ring over ortho and para positions (5 resonance structures). Alkoxide ions (RO^-) have zero resonance stabilization.
- **Substituent Effects on Phenol Acidity:**

$$\begin{aligned}\text{Picric acid } (pK_a = 0.38) > 2,4\text{-dinitrophenol } (4.0) > p\text{-nitrophenol } (7.15)\\ > o\text{-nitrophenol } (7.23) > \text{Phenol } (10.0) > p\text{-cresol } (10.2)\end{aligned} \tag{45}$$

Note: p-nitrophenol is more acidic than o-nitrophenol because the ortho isomer is stabilized by intramolecular hydrogen bonding, reducing proton release!

Kolbe's & Reimer-Tiemann Reactions

1. Kolbe's Reaction (Synthesis of Salicylic Acid): Sodium phenoxide is treated with weak electrophile CO_2 at 400 K and $4-7$ atm, followed by acidification:

$$C_6H_5ONa + CO_2 \xrightarrow[4-7\text{ atm}]{400\text{ K}} \text{Sodium Salicylate} \xrightarrow{H^+} \text{Salicylic Acid (}\mathbf{o}\text{-hydroxybenzoic acid)} \tag{46}$$

Salicylic acid acetylates with acetic anhydride to yield **Aspirin** (acetylsalicylic acid).

2. Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde): Phenol treated with $CHCl_3$ and aqueous NaOH at 340 K:

$$C_6H_5OH + CHCl_3 + 3NaOH \longrightarrow \text{Salicylaldehyde (}\mathbf{o}\text{-hydroxybenzaldehyde)} + 3NaCl + 2H_2O \tag{47}$$

Mechanistic Detail: The electrophile is singlet dichlorocarbene (: CCl_2) generated in situ. Attack occurs predominantly at the ortho position due to cyclic sodium chelation in the transition state. If CCl_4 is used instead of $CHCl_3$, **salicylic acid** is formed directly!

6.3 Ethers: Williamson Synthesis & HI Cleavage

NEET Trap: Substrate Rules for Williamson Ether Synthesis

Williamson ether synthesis involves nucleophilic displacement of an alkyl halide by an alkoxide:

$$\mathrm{R-O^-Na^+ + R'-X \xrightarrow{S_N2} R-O-R' + NaX} \tag{48}$$

The Strict Constraint: The alkyl halide ($\mathrm{R'-X}$) **must be methyl or primary** (1°).

- **Correct Preparation of tert-Butyl Methyl Ether:**

$$\mathrm{(CH_3)_3C-O^-Na^+}\ \text{(sodium } \textit{tert}\text{-butoxide)} + \mathrm{CH_3-I \xrightarrow{S_N2} (CH_3)_3C-O-CH_3 + NaI} \quad (\text{Yield } \approx 95\%) \tag{49}$$

- **Fatal Reversal Trap:** If sodium methoxide is treated with tert-butyl bromide:

$$\mathrm{CH_3-O^-Na^+ + (CH_3)_3C-Br \xrightarrow{E2} CH_2=C(CH_3)_2}\ \text{(2-methylpropene)} + \mathrm{CH_3OH + NaBr} \tag{50}$$

Because tert-butyl bromide is sterically crowded, the strongly basic methoxide ion induces **exclusive** $E2$ **elimination**!

Cleavage of Ethers with Hydroiodic Acid (HI)

When unsymmetrical ether $\mathrm{R-O-R'}$ is cleaved with cold, concentrated HI:

1. **Case 1: Both Alkyl Groups are 1° or 2° (S_N2 Pathway):** The protonated ether $[\mathrm{R-OH^+-R'}]$ is attacked by iodide ion ($\mathrm{I^-}$) at the **less sterically hindered carbon**:

$$\mathrm{CH_3-O-CH_2CH_2CH_3 + HI \xrightarrow{S_N2} CH_3I + CH_3CH_2CH_2OH} \tag{51}$$

2. **Case 2: One Alkyl Group is 3° or Resonance-Stabilized (S_N1 Pathway):** Heterolysis occurs first to produce the stable 3° or benzylic carbocation:

$$\mathrm{(CH_3)_3C-O-CH_3 + HI \xrightarrow{S_N1} (CH_3)_3C-I + CH_3OH} \tag{52}$$

3. **Case 3: Alkyl Aryl Ethers (Anisole $\mathrm{C_6H_5-O-CH_3}$):** The $\mathrm{C_{aromatic}-O}$ bond has partial double bond character due to resonance and cannot be cleaved. Cleavage yields exclusively:

$$\mathrm{C_6H_5-O-CH_3 + HI \longrightarrow C_6H_5OH}\ \text{(phenol)} + \mathrm{CH_3I} \tag{53}$$

Note: Excess hot HI *converts all aliphatic alcohols to alkyl iodides:* $\mathrm{R-OH + HI \rightarrow R-I + H_2O}$. *Phenol never converts to iodobenzene!*

7 Aldehydes, Ketones & Carboxylic Acids

7.1 Nucleophilic Addition Mechanisms

Carbonyl Electrophilicity & Relative Reactivity Order

The carbonyl carbon is sp^2-hybridized with planar geometry (120°). Strong electronegativity of oxygen ($\chi = 3.44$ vs $\chi_C = 2.55$) creates a permanent dipole:

$$\mathrm{>C^{\delta+}=O^{\delta-} \longleftrightarrow > \overset{+}{C}-\overset{-}{O}} \tag{54}$$

Nucleophiles attack perpendicular to the carbonyl plane (107° Bürgi-Dunitz angle), transforming the planar sp^2 carbon into a tetrahedral sp^3 alkoxide intermediate. **Reactivity Hierarchy Toward Nucleophiles:**

$$\mathrm{HCHO > CH_3CHO > CH_3COCH_3 > C_6H_5CHO > C_6H_5COCH_3 > C_6H_5COC_6H_5} \tag{55}$$

Governing Factors:

- **Steric Effect:** Bulky alkyl or aryl groups crowd the transition state, hindering nucleophilic approach.
- **Electronic Effect:** Electron-donating $+I$ alkyl groups reduce partial positive charge on the carbonyl carbon. $+M$ resonance from aromatic rings strongly stabilizes the ground state, lowering electrophilicity.

Standard Nucleophilic Addition Reactions

1. **Addition of HCN (Cyanohydrin Formation):** Catalyzed by base (OH^- generates active nucleophile CN^-). Cyanohydrins hydrolyze in acid to α-hydroxy acids:

$$CH_3CHO + HCN \xrightarrow{OH^-} CH_3CH(OH)CN \xrightarrow{H_3O^+} CH_3CH(OH)COOH \text{ (Lactic Acid)} \quad (56)$$

2. **Addition of Sodium Bisulfite ($NaHSO_3$):** Aldehydes and methyl ketones form crystalline, water-soluble bisulfite adducts. The adduct regenerates pure carbonyl on treatment with dilute mineral acid or base (used for industrial separation and purification).
3. **Addition of Alcohols (Acetal/Ketal Formation):** Carbonyls react with 1 equivalent of alcohol in dry HCl to form hemiacetals, and with 2 equivalents to form acetals. Acetals are stable to bases, oxidants, and reducing agents, acting as universal carbonyl protecting groups.
4. **Nucleophilic Addition-Elimination with Ammonia Derivatives ($NH_2 - G$):**

$$> C = O + H_2N - G \xrightarrow{pH4.5} [> C(OH) - NH - G] \xrightarrow{-H_2O} > C = N - G \quad (57)$$

   - Hydroxylamine (NH_2OH) $\rightarrow$ **Oxime** ($> C = N - OH$).
   - Hydrazine (NH_2NH_2) $\rightarrow$ **Hydrazone** ($> C = N - NH_2$).
   - 2,4-Dinitrophenylhydrazine (2,4-DNP / Brady's Reagent) $\rightarrow$ **Yellow/Orange precipitate** (used to detect aldehydes/ketones).
   - Semicarbazide ($NH_2 - CO - NHNH_2$) $\rightarrow$ **Semicarbazone**. (*Trap: Only the hydrazine* $-NH_2$ *is nucleophilic; the amide* $-CONH_2$ *nitrogen lone pair is delocalized into the carbonyl!*)

7.2 Reactions Involving α-Hydrogen: Aldol, Cannizzaro & Haloform

Step-by-Step Mechanism: Aldol Condensation

Aldehydes or ketones possessing at least one α-hydrogen react in dilute alkali (dil. NaOH):

1. **Step 1: Enolate Generation (Acid-Base Equilibrium):**

$$CH_3 - CHO + OH^- \rightleftharpoons [:\bar{C}H_2 - CHO \longleftrightarrow CH_2 = CH - O^-] + H_2O \quad (58)$$

2. **Step 2: Nucleophilic Attack on Second Carbonyl Molecule (RDS):**

$$CH_3 - CHO + :\bar{C}H_2 - CHO \longrightarrow CH_3 - CH(O^-) - CH_2 - CHO \quad (59)$$

3. **Step 3: Protonation to Aldol (β-Hydroxyaldehyde):**

$$CH_3 - CH(O^-) - CH_2 - CHO + H_2O \longrightarrow CH_3 - CH(OH) - CH_2 - CHO \text{ (3-hydroxybutanal)} + OH^- \quad (60)$$

4. **Step 4: Dehydration on Heating ($E1cB$ Mechanism):** Deprotonation of the acidic α-carbon creates an enolate, followed by expulsion of hydroxide ion (OH^-) to yield a conjugated α,β-unsaturated aldehyde:

$$CH_3 - CH(OH) - CH_2 - CHO \xrightarrow{\Delta, -H_2O} CH_3 - CH = CH - CHO \text{ (But-2-enal / Crotonaldehyde)} \quad (61)$$

Step-by-Step Mechanism: Cannizzaro Reaction

Aldehydes **lacking α-hydrogen** ($HCHO, C_6H_5CHO, (CH_3)_3CCHO$) disproportionate in concentrated alkali (50% NaOH):

1. **Step 1: Reversible Addition of Hydroxide:**

$$C_6H_5CHO + OH^- \rightleftharpoons C_6H_5 - CH(OH) - O^- \quad \text{(Monoanion)} \quad (62)$$

2. **Step 2: Rate-Determining Hydride Shift (H^-):** The alkoxide reforms the carbonyl double bond, transferring a hydride ion directly to a second aldehyde molecule:

$$C_6H_5 - CH(O^-) - OH + C_6H_5CHO \xrightarrow{\text{slow, RDS}} C_6H_5COOH + C_6H_5CH_2O^- \quad (63)$$

3. **Step 3: Fast Acid-Base Proton Exchange:**

$$C_6H_5COOH + C_6H_5CH_2O^- \xrightarrow{\text{fast}} C_6H_5COO^-Na^+ \text{ (Sodium Benzoate)} + C_6H_5CH_2OH \text{ (Benzyl Alcohol)} \quad (64)$$

Cross-Cannizzaro Reaction: When benzaldehyde reacts with formaldehyde in concentrated alkali, formaldehyde is oxidized exclusively to sodium formate (HCOONa) while benzaldehyde is reduced to benzyl alcohol:

$$C_6H_5CHO + HCHO + NaOH \longrightarrow C_6H_5CH_2OH + HCOONa \quad (65)$$

Cause: Formaldehyde is sterically uncrowded and far more electrophilic than benzaldehyde, undergoing nucleophilic attack by OH^- first!

The Haloform (Iodoform) Reaction Test

Structural Requirement: Reagent I_2/NaOH (or NaOI) oxidizes and cleaves compounds containing:

1. $CH_3 - C = O$ (Methyl ketone group, e.g., acetaldehyde, acetone, acetophenone).
2. $CH_3 - CH(OH) -$ (Methyl carbinol group oxidizable in situ to methyl ketone, e.g., ethanol, propan-2-ol, butan-2-ol).

$$R - CO - CH_3 + 3I_2 + 4NaOH \longrightarrow CHI_3 \downarrow \text{ (Yellow precipitate, m.p. } 119^\circ C) + R - COONa + 3NaI + 3H_2O \quad (66)$$

Classical NEET Distinction Traps:

- **Ethanol vs Methanol:** Ethanol gives positive iodoform test; methanol gives negative test.
- **Pentan-2-one vs Pentan-3-one:** Pentan-2-one is a methyl ketone (positive); pentan-3-one is not (negative).
- **Acetaldehyde vs Benzaldehyde:** Acetaldehyde gives yellow iodoform; benzaldehyde does not.

7.3 Distinction Tests & Reductions

Distinction Tests for Carbonyl Functions

- **Tollens' Test (Silver Mirror Test):** Tollens' reagent ($[Ag(NH_3)_2]^+OH^-$) is reduced by **all aldehydes** (aliphatic and aromatic) and α-hydroxy ketones to a shining silver mirror (Ag $\downarrow$):

$$R - CHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow R - COO^- + 2Ag \downarrow + 4NH_3 + 2H_2O \quad (67)$$

- **Fehling's Test:** Alkaline cupric tartrate complex (Cu^{2+}) is reduced to a **red precipitate of cuprous oxide** ($Cu_2O \downarrow$) by **aliphatic aldehydes only**:

$$R - CHO + 2Cu^{2+} + 5OH^- \longrightarrow R - COO^- + Cu_2O \downarrow \text{(red)} + 3H_2O \quad (68)$$

NEET Trap: Benzaldehyde and all aromatic aldehydes FAIL Fehling's test because aromatic resonance stabilizes the carbonyl, making it an insufficiently strong reducing agent to reduce the tartrate-chelated Cu^{2+} ion!

Clemmensen vs Wolff-Kishner Carbonyl Reduction

- **Clemmensen Reduction:** Uses amalgamated zinc and concentrated hydrochloric acid ($Zn - Hg$/conc. HCl):

$$> C = O + 4[H] \xrightarrow{Zn-Hg/\text{conc. HCl}} > CH_2 + H_2O \quad (69)$$

Constraint: Cannot be used for substrates containing acid-sensitive groups ($-OH, -OR, -C = C-$), which undergo elimination or addition in strong acid.

- **Wolff-Kishner Reduction:** Carbonyl converted to hydrazone, then heated with KOH in ethylene glycol

at $453 - 473$ K:

$$> C = O + NH_2NH_2 \xrightarrow{-H_2O} > C = N - NH_2 \xrightarrow[\text{glycol},\Delta]{KOH} > CH_2 + N_2 \uparrow \quad (70)$$

Constraint: Cannot be used for substrates containing base-sensitive groups (alkyl halides, esters), which undergo nucleophilic displacement or saponification.

7.4 Carboxylic Acids & Derivatives

Resonance in Carboxylic Acids & The Ortho Effect

1. Resonance of Carboxylate Anion: Deprotonation of carboxylic acid yields the carboxylate anion ($R - COO^-$), which is stabilized by two **completely equivalent** resonance structures sharing negative charge equally between two electronegative oxygen atoms ($C - O$ bond lengths identical at 1.26 Å). This makes carboxylic acids vastly more acidic than phenols.
2. The Ortho Effect in Benzoic Acids: Any substituent at the **ortho position** of benzoic acid—regardless of whether it is electron-withdrawing ($-NO_2$) or electron-releasing ($-CH_3, -OCH_3$)—increases the acidic strength compared to benzoic acid:

$$o\text{-toluic acid } (pK_a = 3.91) > \text{benzoic acid } (pK_a = 4.20) > m\text{-toluic acid } (4.27) > p\text{-toluic acid } (4.37) \quad (71)$$

Physical Cause: Steric crowding by the ortho substituent twists the $-COOH$ group out of the plane of the benzene ring. This **steric inhibition of resonance (SIR)** eliminates cross-conjugation between the ring and the carbonyl, allowing unhindered resonance delocalization of negative charge across the two carboxylate oxygens!

Hell-Volhard-Zelinsky (HVZ) Reaction & Decarboxylation

1. HVZ Reaction (α-Halogenation): Carboxylic acids possessing α-hydrogen react with chlorine or bromine in the presence of catalytic red phosphorus:

$$R - CH_2 - COOH + X_2 \xrightarrow{\text{red P}} R - CH(X) - COX \xrightarrow{H_2O} R - CH(X) - COOH\ (\alpha\text{-halocarboxylic acid}) + HX \quad (72)$$

Note: Formic acid ($HCOOH$*) and benzoic acid (*C_6H_5COOH*) fail the HVZ reaction due to absence of α-hydrogens!*
2. Soda-Lime Decarboxylation: Sodium carboxylates heated with soda lime ($NaOH + CaO$ in $3:1$ mass ratio) undergo loss of carbon dioxide:

$$R - COONa + NaOH \xrightarrow[\Delta]{CaO} R - H + Na_2CO_3 \quad (73)$$

CaO acts as a desiccant, keeping NaOH dry and raising the reaction fusion temperature.

8 Nitrogen-Containing Compounds: Amines & Diazonium Chemistry

8.1 Basicity of Amines: Gas Phase vs Aqueous Phase

Thermodynamic Factors Governing Amine Basicity

The basicity of amines is measured by K_b or pK_b ($pK_b = -\log K_b$; lower pK_b indicates stronger base). Basicity reflects the thermodynamic stability of the conjugate ammonium cation (RNH_3^+). **1. Gas Phase (Isolated Molecules):** In the gas phase, solvent molecules are absent. Basicity is governed strictly by the electron-donating $+I$ inductive effect of alkyl groups, which stabilizes the positive charge on nitrogen:

$$\text{Gas Phase Basicity: } 3^\circ\ R_3N > 2^\circ\ R_2NH > 1^\circ\ RNH_2 > NH_3 \quad (74)$$

2. Aqueous Solution (Interplay of Three Opposing Factors): In water, three thermodynamic effects

operate concurrently:

1. **Inductive Effect ($+I$):** Stabilizes cation in order $3^\circ > 2^\circ > 1^\circ > NH_3$.
2. **Hydration Energy (H-Bonding):** The conjugate ammonium ion forms hydrogen bonds with water molecules. Greater number of hydrogens attached to positive nitrogen yields higher hydration energy: RNH_3^+ (3 H-bonds) > $R_2NH_2^+$ (2 H-bonds) > R_3NH^+ (1 H-bond). Hydration stabilizes in order: $NH_4^+ > 1^\circ > 2^\circ > 3^\circ$.
3. **Steric Hindrance to Protonation:** Bulky alkyl groups shield the nitrogen lone pair, impeding proton attack in order: $1^\circ > 2^\circ > 3^\circ$.

Authoritative Aqueous Basicity Orders (NEET Core Benchmark)

The combined net effect produces the following experimentally established basicity sequences in aqueous medium:

1. **Methyl-Substituted Amines ($2^\circ > 1^\circ > 3^\circ > NH_3$, Mnemonic: "213"):**

$$(CH_3)_2NH\ (pK_b = 3.27) > CH_3NH_2\ (3.38) > (CH_3)_3N\ (4.22) > NH_3\ (4.75) \tag{75}$$

2. **Ethyl-Substituted Amines ($2^\circ > 3^\circ > 1^\circ > NH_3$, Mnemonic: "231"):**

$$(C_2H_5)_2NH\ (pK_b = 3.00) > (C_2H_5)_3N\ (3.25) > C_2H_5NH_2\ (3.29) > NH_3\ (4.75) \tag{76}$$

3. **Arylamines (Aniline):** Aniline ($C_6H_5NH_2$, $pK_b = 9.38$) is vastly weaker than ammonia ($pK_b = 4.75$) because its nitrogen lone pair is delocalized over the ortho and para positions of the aromatic ring (5 canonical forms). The anilinium cation ($C_6H_5NH_3^+$) has only 2 resonance structures, making protonation thermodynamically disfavored!

8.2 Gabriel Phthalimide & Hoffmann Bromamide Mechanisms

Gabriel Phthalimide Synthesis: Pure Primary Aliphatic Amines

Gabriel synthesis produces pure 1° aliphatic amines free from secondary or tertiary contamination:

1. **Potassium Salt Formation:** Phthalimide is deprotonated by ethanolic KOH to form potassium phthalimide (the imide hydrogen is acidic due to two flanking carbonyls).
2. **Alkylation (S_N2):** Nucleophilic attack of the phthalimide anion on a primary alkyl halide ($R-X$) yields N-alkylphthalimide.
3. **Hydrazinolysis / Alkaline Hydrolysis:** Alkaline hydrolysis with NaOH yields pure primary amine and sodium phthalate:

$$\begin{aligned}\text{Phthalimide} \xrightarrow{\text{KOH}} \text{Potassium Phthalimide} \xrightarrow{\text{R}-\text{X}} N\text{-Alkylphthalimide} \\ \xrightarrow{\text{NaOH(aq)}} R-NH_2 + \text{Sodium Phthalate}\end{aligned} \tag{77}$$

NEET Trap: Why Aromatic Amines CANNOT be Prepared by Gabriel Synthesis

Aniline ($C_6H_5NH_2$) cannot be prepared by Gabriel phthalimide synthesis because the required aryl halide (C_6H_5-X) does not undergo nucleophilic aromatic substitution (S_N2) with the phthalimide anion under standard conditions due to partial double bond character of the $C-X$ bond and steric/electronic repulsion of the aromatic ring!

Step-by-Step Mechanism: Hoffmann Bromamide Degradation

Primary acid amides react with bromine and aqueous alkali ($Br_2 + 4KOH$) to yield primary amines with **one less carbon atom**:

$$R-CONH_2 + Br_2 + 4KOH \longrightarrow R-NH_2 + K_2CO_3 + 2KBr + 2H_2O \tag{78}$$

Elementary Mechanistic Steps:

1. **Formation of N-Bromoamide:** $RCONH_2 + OH^- \rightarrow RCONH^- \xrightarrow{Br_2} R-CONHBr + Br^-$.

2. **Formation of Bromoamide Anion:** $\mathrm{R-CONHBr + OH^- \rightleftharpoons [R-CON-Br]^- + H_2O}$.
3. **Rate-Determining Concerted 1,2-Shift (Isocyanate Formation):** Migration of alkyl/aryl group R with its electron pair from carbonyl carbon to electron-deficient nitrogen with simultaneous loss of bromide ion:

$$\mathrm{[R-CO-\bar{N}-Br] \xrightarrow{slow,\ RDS} R-N=C=O\ (Alkyl\ Isocyanate) + Br^-} \tag{79}$$

*Stereochemical Note: The migrating group R migrates with **complete retention of configuration!** If chiral, optical activity is preserved.*

4. **Alkaline Hydrolysis of Isocyanate:**

$$\mathrm{R-N=C=O + 2OH^- \longrightarrow R-NH_2 + CO_3^{2-}} \tag{80}$$

8.3 Distinction Tests for Amines

Hinsberg, Carbylamine & Nitrous Acid Tests

1. **Hinsberg Test (Benzenesulfonyl Chloride $\mathrm{C_6H_5SO_2Cl}$):**
   - 1° **Amines:** React to form N-alkylbenzenesulfonamide ($\mathrm{C_6H_5SO_2NHR}$). The remaining hydrogen on nitrogen is strongly acidic due to the electron-withdrawing $-\mathrm{SO_2}-$ group $\implies$ **Dissolves completely in aqueous** KOH.
   - 2° **Amines:** React to form N,N-dialkylbenzenesulfonamide ($\mathrm{C_6H_5SO_2NR_2}$). Lacks acidic hydrogen on nitrogen $\implies$ **Insoluble in aqueous** KOH.
   - 3° **Amines:** Possess no hydrogen on nitrogen $\implies$ **Does not react at all**.
2. **Carbylamine Test (Isocyanide Test):** Exclusive to **primary amines** (both aliphatic and aromatic). Heated with chloroform and alcoholic KOH:

$$\mathrm{R-NH_2 + CHCl_3 + 3KOH \xrightarrow{\Delta} R-N\equiv C\ (Carbylamine,\ foul\ odor) + 3KCl + 3H_2O} \tag{81}$$

Secondary and tertiary amines fail the carbylamine test!

3. **Reaction with Nitrous Acid ($\mathrm{HNO_2 = NaNO_2 + HCl}$):**
   - 1° **Aliphatic Amines:** Yield unstable aliphatic diazonium salts that decompose quantitatively to alcohols with quantitative evolution of nitrogen gas: $\mathrm{RNH_2 + HNO_2 \rightarrow ROH + N_2 \uparrow + H_2O}$.
   - 1° **Aromatic Amines (Aniline):** Form stable benzenediazonium chloride at ice-cold temperature ($0-5°\mathrm{C}$).
   - 2° **Amines (Aliphatic & Aromatic):** React to form yellow oily N-nitrosamines ($\mathrm{R_2N-NO}$), which give **Liebermann's nitroso test**.
   - 3° **Aliphatic Amines:** Form water-soluble trialkylammonium nitrite salts ($\mathrm{[R_3NH]^+NO_2^-}$).

8.4 Benzenediazonium Chemistry & Synthetic Transformations

Sandmeyer, Gattermann, Balz-Schiemann & Coupling Reactions

Benzenediazonium chloride ($\mathrm{C_6H_5N_2^+Cl^-}$) is prepared by diazotization of aniline with $\mathrm{NaNO_2 + 2HCl}$ at $273-278$ K ($0-5°\mathrm{C}$). It undergoes two major classes of reactions: **1. Replacement Reactions (Loss of $\mathrm{N_2}\uparrow$):**

- **Sandmeyer Reaction:** $\mathrm{Cu_2Cl_2/HCl \rightarrow C_6H_5Cl}$; $\mathrm{Cu_2Br_2/HBr \rightarrow C_6H_5Br}$; $\mathrm{CuCN/KCN \rightarrow C_6H_5CN}$ (higher yields via cuprous salts).
- **Gattermann Reaction:** Cu powder/HCl $\rightarrow \mathrm{C_6H_5Cl}$; Cu powder/HBr $\rightarrow \mathrm{C_6H_5Br}$.
- **Balz-Schiemann Reaction:** $\mathrm{HBF_4 \rightarrow C_6H_5N_2^+BF_4^- \xrightarrow{\Delta} C_6H_5F + BF_3 + N_2 \uparrow}$ (best method for fluorobenzene).
- **Replacement by Iodine:** Warm with aqueous KI $\rightarrow \mathrm{C_6H_5I + KCl + N_2 \uparrow}$ (no copper catalyst needed).

- **Deamination (Replacement by H):** Reduction with hypophosphorous acid (H_3PO_2/H_2O) or ethanol (CH_3CH_2OH):

$$C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \xrightarrow{Cu^+} C_6H_6 + H_3PO_3 + HCl + N_2\uparrow \tag{82}$$

2. Azo Coupling Reactions (Retention of Diazo Linkage $-N=N-$): The diazonium ion acts as a weak electrophile, attacking highly activated aromatic rings at the para position:

- **Coupling with Phenol (Alkaline Medium, pH $9-10$):** Forms **orange azo dye** (*p*-hydroxyazobenzene):

$$C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{pH9-10} C_6H_5-N=N-C_6H_4-OH\ (\text{Orange Dye}) + HCl \tag{83}$$

- **Coupling with Aniline (Mildly Acidic Medium, pH $4-5$):** Forms **yellow azo dye** (*p*-aminoazobenzene):

$$C_6H_5N_2^+Cl^- + C_6H_5NH_2 \xrightarrow{pH4-5} C_6H_5-N=N-C_6H_4-NH_2\ (\text{Yellow Dye}) + HCl \tag{84}$$

9 Biomolecules (NEET Unit 19)

9.1 Carbohydrates: Classification & Stereochemistry

Definition & D/L Stereochemical Nomenclature

Carbohydrates are optically active polyhydroxy aldehydes or polyhydroxy ketones, or substances that produce them upon hydrolysis.

- **D/L Configuration (Rosanoff Convention):** Assigned by comparing the configuration of the highest-numbered chiral carbon (the penultimate asymmetric carbon, C_5 in hexoses) in a vertical Fischer projection to **D-glyceraldehyde**:
  - **D-Sugar:** The $-OH$ group at the lowest asymmetric center projects to the **right**.
  - **L-Sugar:** The $-OH$ group projects to the **left**.
- *Crucial Principle: D and L designate configurational stereochemistry only. They have ZERO correlation to experimental dextrorotation $(+)$ or levorotation $(-)$. D-glucose is dextrorotatory, whereas D-fructose is strongly levorotatory!*

Structure Elucidation of D-(+)-Glucose

Experimental evidence establishing the open-chain structure of D-glucose:

1. **Molecular Formula:** $C_6H_{12}O_6$.
2. **Straight Chain of 6 Carbons:** Prolonged heating with hydroiodic acid and red phosphorus yields **n-hexane**:

$$C_6H_{12}O_6 \xrightarrow[\Delta]{HI/red\ P} CH_3-(CH_2)_4-CH_3\ (n\text{-hexane}) \tag{85}$$

3. **Carbonyl Group ($>C=O$):** Reacts with hydroxylamine (NH_2OH) to form an oxime, and with HCN to form a cyanohydrin.
4. **Aldehyde Function ($-CHO$):** Oxidation with mild bromine water (Br_2/H_2O) selectively oxidizes glucose to **gluconic acid** ($HOOC-(CHOH)_4-CH_2OH$). (Ketones are not oxidized by bromine water).
5. **Five Hydroxyl Groups ($-OH$):** Acetylation with acetic anhydride (Ac_2O) yields **glucose pentaacetate**. Because it is stable, all five $-OH$ groups reside on separate carbon atoms.
6. **Primary Alcohol at C_6 ($-CH_2OH$):** Oxidation with concentrated nitric acid (HNO_3) oxidizes both the terminal aldehyde and the primary alcohol to dicarboxylic **saccharic acid (glucaric acid)**:

$$CHO-(CHOH)_4-CH_2OH \xrightarrow{HNO_3} HOOC-(CHOH)_4-COOH\ (\text{Saccharic Acid}) \tag{86}$$

9.2 Cyclic Pyranose Structures, Anomers & Mutarotation

Failure of Open-Chain Structure & The Cyclic Pyranose Ring

1. Evidence Opposing the Simple Open-Chain Aldehyde:

- Glucose does not react with $NaHSO_3$, does not form 2,4-DNP adduct, and does not restore the pink color of Schiff's reagent.
- Glucose penta-acetate does not react with hydroxylamine (proves absence of a free $-CHO$ group).
- Glucose exists in two distinct crystalline stereoisomers (α and β) with different physical properties.

2. Cyclic Hemiacetal Formation: Intramolecular nucleophilic attack of the C_5-OH group onto the C_1 carbonyl carbon produces a 6-membered **pyranose ring**. The C_1 carbon becomes an asymmetric center known as the **anomeric carbon**:

- **α-D-Glucopyranose:** In Haworth projection, the anomeric $-OH$ at C_1 points **downward** (*trans* to the $-CH_2OH$ group at C_6). Melting point 146°C, specific rotation $[\alpha]_D = +112°$.
- **β-D-Glucopyranose:** The anomeric $-OH$ at C_1 points **upward** (*cis* to the $-CH_2OH$ group at C_6). In the chair conformation, all bulky groups ($-OH$ and $-CH_2OH$) occupy equatorial positions $\implies$ thermodynamically more stable. Melting point 150°C, specific rotation $[\alpha]_D = +18.7°$.

3. Mutarotation Mechanism: When pure crystals of either α-D-glucose (+112°) or β-D-glucose (+18.7°) are dissolved in water, the optical rotation spontaneously drifts over time until reaching an equilibrium value of +52.7°:

$$\alpha\text{-D-glucose } (+112°) \underset{\approx 36\%}{\overset{H_2O}{\rightleftharpoons}} \text{Open-Chain Form } (0.02\%) \underset{\approx 64\%}{\overset{H_2O}{\rightleftharpoons}} \beta\text{-D-glucose } (+18.7°) \tag{87}$$

This reversible interconversion between anomers via the open-chain aldehyde is called **mutarotation**.

9.3 Disaccharides, Glycosidic Linkages & Polysaccharides

Disaccharide Chemistry: Linkages & Reducing Character

Sugar	Monomer Units	Glycosidic Bond	Free Anomeric C?	Reducing?
Sucrose (Cane sugar)	α-D-glucose + β-D-fructose	$\alpha(\mathbf{1 \to 2})\beta$	None (C_1 of Glc and C_2 of Fru tied up)	**Non-Reducing**
Maltose (Malt sugar)	α-D-glucose + α-D-glucose	$\alpha(\mathbf{1 \to 4})$	Yes (C_1 of second glucose free)	**Reducing**
Lactose (Milk sugar)	β-D-galactose + β-D-glucose	$\beta(\mathbf{1 \to 4})$	Yes (C_1 of glucose free)	**Reducing**

Invert Sugar & The Optical Inversion of Sucrose

Sucrose is **dextrorotatory** ($[\alpha]_D = +66.5°$). Upon acid hydrolysis or enzymatic cleavage by *invertase*, it yields an equimolar mixture of D-(+)-glucose and D-(-)-fructose:

$$C_{12}H_{22}O_{11} \text{ (Sucrose, } +66.5°) + H_2O \xrightarrow{H^+} \text{D-(+)-glucose } (+52.7°) + \text{D-(-)-fructose } (-92.4°) \tag{88}$$

Because the levorotation of fructose (−92.4°) is numerically far greater than the dextrorotation of glucose (+52.7°), the net resulting hydrolyzed mixture is **levorotatory** ($[\alpha]_{net} \approx -39.7°$).
Definition: This reversal in the sign of optical rotation from dextro (+) to levo (−) is called the **inversion of cane sugar**, and the resulting equimolar mixture is termed **invert sugar** (found naturally in honey).

Polysaccharides: Starch, Cellulose & Glycogen

- **Starch (Plant Storage Polysaccharide):** Polymer of α-D-glucose consisting of two distinct structural components:
  1. **Amylose ($15-20\%$):** Water-soluble, linear unbranched chain of $200-1000$ α-D-glucose units linked

by $\alpha(1 \to 4)$ **glycosidic bonds**. Forms a hollow helical coil that entraps triiodide ions (I_3^-) to produce a characteristic deep blue-black color.

2. **Amylopectin ($80-85\%$):** Water-insoluble, highly branched polymer. Linear chains contain $\alpha(1 \to 4)$ linkages; branching points occur every $24-30$ units via $\alpha(1 \to 6)$ **glycosidic bonds**.

- **Cellulose (Plant Cell Wall Structural Polymer):** Unbranched, linear polymer composed exclusively of β-D-glucose units joined by $\beta(1 \to 4)$ **glycosidic bonds**. Chains lie parallel, held together by massive intermolecular hydrogen bonds to form rigid microfibrils. Humans cannot digest cellulose due to lack of the enzyme cellulase.
- **Glycogen (Animal Starch):** Stored in mammalian liver and muscle. Structurally similar to amylopectin, but possesses far more frequent branching ($\alpha(1 \to 6)$ branches every $8-12$ glucose residues).

9.4 Amino Acids, Zwitterions & The Isoelectric Point (pI)

Structure, Stereochemistry & Essential Amino Acids

Amino acids contain both an amino group ($-NH_2$) and a carboxyl group ($-COOH$):

$$R-CH(NH_2)-COOH \quad (\alpha\text{-amino acid}) \tag{89}$$

- **Chirality Exception:** All 20 standard proteinogenic amino acids are chiral and optically active **except Glycine** (H_2N-CH_2-COOH), where $R=H$, which is achiral and optically inactive!
- **L-Configuration:** All natural amino acids in proteins belong to the L-series (the $-NH_2$ group resides on the left in Fischer projections).
- **Essential Amino Acids (Cannot be synthesized in human body; must be supplied in diet):** Valine, Leucine, Isoleucine, Phenylalanine, Tryptophan, Threonine, Methionine, Lysine (plus Arginine and Histidine, semi-essential in infants). *(Mnemonic: PVT TIM HaLL).*

Zwitterion Structure & Formulation of Isoelectric Point (pI)

In aqueous solution, the acidic carboxyl group transfers a proton to the basic amino group, generating a dipolar internal salt known as a **zwitterion**:

$$R-CH(NH_2)-COOH \rightleftharpoons R-CH(NH_3^+)-COO^- \quad (\text{Zwitterion: net charge } = 0) \tag{90}$$

1. Behavior in Varying pH Media:

- In acidic solution ($pH < pI$): Protonation gives cation $R-CH(NH_3^+)-COOH \implies$ migrates toward **cathode** in electric field.
- In basic solution ($pH > pI$): Deprotonation gives anion $R-CH(NH_2)-COO^- \implies$ migrates toward **anode** in electric field.

2. Isoelectric Point (pI): The exact pH at which the amino acid carries no net electrical charge and does not migrate in an electric field:

$$pI = \frac{pK_{a1}(\alpha\text{–COOH}) + pK_{a2}(\alpha\text{–}NH_3^+)}{2} \quad (\text{for neutral amino acids}) \tag{91}$$

*Crucial Property: At its isoelectric point (pI), an amino acid exhibits **minimum aqueous solubility**, permitting selective precipitation and separation!*

9.5 Peptide Bond Resonance & Structural Hierarchy of Proteins

The Peptide Bond (-CO – NH–) & Planarity

A peptide bond is an amide linkage formed by condensation of the $-COOH$ group of one amino acid with

the $-NH_2$ group of another with loss of water:

$$\begin{aligned} &R-CH(NH_2)-COOH + H_2N-CH(R')-COOH \\ &\xrightarrow{-H_2O} R-CH(NH_2)-\mathbf{CO}-\mathbf{NH}-CH(R')-COOH \end{aligned} \tag{92}$$

Resonance Stabilization of the Peptide Bond: Delocalization of the nitrogen lone pair into the carbonyl group confers $\approx 40\%$ **partial double bond character** to the $C-N$ bond:

$$-C(=O)-NH- \longleftrightarrow -C(O^-)=\overset{+}{N}H- \tag{93}$$

Consequences:

1. The $C-N$ bond length (1.32 Å) is substantially shorter than a normal $C-N$ single bond (1.47 Å).
2. Free rotation about the $C-N$ bond is restricted; the six atoms of the peptide group ($C_\alpha-CO-NH-C_\alpha$) are strictly coplanar.
3. The *trans*-conformation is overwhelmingly favored to minimize steric clash between adjacent side chains (R).

The Four Structural Levels of Protein Organization

1. **Primary Structure (1°):** The linear sequence of amino acids linked covalently by peptide bonds. Any alteration in sequence alters biological function (e.g., in sickle-cell anemia, glutamate at position 6 of the β-chain of hemoglobin is mutated to valine).
2. **Secondary Structure (2°):** Regular folding of the polypeptide backbone stabilized exclusively by **hydrogen bonds** between $>C=O$ and $-NH-$ groups:
   - **α-Helix:** Polypeptide coils into a right-handed screw. Stabilized by intramolecular hydrogen bonding between the $>C=O$ of residue n and the $-NH-$ of residue $(n+4)$. Each turn contains 3.6 amino acids with pitch 5.4 Å. Proline acts as a helix breaker.
   - **β-Pleated Sheet:** Polypeptide chains are stretched side-by-side in zigzag sheets, held by intermolecular hydrogen bonds. Can be parallel or antiparallel (e.g., silk fibroin).
3. **Tertiary Structure (3°):** Overall 3D spatial folding of the polypeptide chain into compact shapes:
   - **Fibrous Proteins:** Linear, insoluble in water, structural role (keratin in hair/nails, collagen in tendons, myosin in muscle).
   - **Globular Proteins:** Spherical, water-soluble, metabolic role (enzymes, hemoglobin, albumin).
   - **Stabilizing Forces:** Hydrogen bonds, **disulfide bonds ($-S-S-$ between cysteine residues, covalent)**, ionic/salt bridges ($-COO^- \cdots NH_3^+$), hydrophobic interactions, and van der Waals forces.
4. **Quaternary Structure (4°):** Spatial arrangement of two or more independent tertiary polypeptide subunits into an oligomeric protein (e.g., adult Hemoglobin is a tetramer consisting of 2α and 2β subunits).

Denaturation of Proteins: What Survives and What Is Destroyed?

When a native protein is exposed to physical changes (heating, radiation) or chemical changes (extreme pH, urea, organic solvents, heavy metals):

- **Destroyed:** Hydrogen bonds, salt bridges, and hydrophobic interactions rupture. The delicate $2^\circ, 3^\circ$, and 4° spatial architectures unfold. Globular proteins precipitate, and biological activity is irreversibly lost.
- **Retained Intact: The primary structure (1°) remains 100% completely intact!** Covalent peptide bonds ($-CO-NH-$) are not cleaved under normal denaturation conditions.
- **Classic NCERT Examples:** Coagulation of egg white (albumin) upon boiling; curdling of milk caused by lactic acid produced by bacteria.

9.6 Nucleic Acids: DNA & RNA Architecture

Chemical Composition: Sugars, Bases & Nucleotides

Nucleic acids are unbranched biopolymers of nucleotides:

1. **Pentose Sugar:** β-D-ribose in RNA; β-D-2-deoxyribose in DNA (lacks oxygen atom at C_2' position).
2. **Nitrogenous Bases:**
   - **Purines (Bicyclic 9-membered):** Adenine (A) and Guanine (G) in both DNA and RNA.
   - **Pyrimidines (Monocyclic 6-membered):** Cytosine (C) and Thymine (T, 5-methyluracil) in DNA; Cytosine (C) and Uracil (U) in RNA.
3. **Nucleoside vs Nucleotide:**
   - Nucleoside = Pentose Sugar + Base (joined by N-glycosidic linkage between C_1' of sugar and N_9 of purine or N_1 of pyrimidine).
   - Nucleotide = Nucleoside + Phosphate (esterified at C_5' of pentose sugar).
4. **Phosphodiester Linkage:** Nucleotides polymerize via $3', 5'$**-phosphodiester bonds** between the $3' - \text{OH}$ of one nucleotide sugar and the $5' - \text{OH}$ of the next nucleotide sugar, establishing directional polarity ($5' \to 3'$).

Watson-Crick Double Helix Model & Chargaff's Rules

James Watson and Francis Crick (1953) established the B-DNA double helix structure:

1. **Antiparallel Strands:** Two polynucleotide chains wound around a central axis in a right-handed helix. One strand runs $5' \to 3'$, and the complementary strand runs $3' \to 5'$.
2. **Complementary Base Pairing (Hydrogen Bonds):**

$$\mathbf{A = T} \quad \text{(Adenine pairs with Thymine via 2 Hydrogen Bonds)} \tag{94}$$

$$\mathbf{G \equiv C} \quad \text{(Guanine pairs with Cytosine via 3 Hydrogen Bonds)} \tag{95}$$

   Because $G \equiv C$ pairs have 3 hydrogen bonds, DNA with high GC content has a higher melting temperature (T_m).
3. **Geometric Dimensions of B-DNA:**
   - Helix Diameter: 2.0 nm (20 Å).
   - Pitch of Helix (one complete 360° turn): 3.4 nm (34 Å).
   - Base Pairs per Turn: 10 base pairs.
   - Distance between Adjacent Base Pairs: 0.34 nm (3.4 Å).
4. **Chargaff's Rules (Applies Exclusively to Double-Stranded DNA):**

$$[\text{A}] = [\text{T}], \quad [\text{G}] = [\text{C}] \implies [\text{A} + \text{G}] = [\text{T} + \text{C}] \quad \left(\frac{\text{Total Purines}}{\text{Total Pyrimidines}} = 1.0 \right) \tag{96}$$

   Note: The ratio $\frac{\text{A}+\text{T}}{\text{G}+\text{C}}$ *is not necessarily equal to 1; it is variable and species-specific.*

9.7 Vitamins & Hormones

Comprehensive Vitamin Classification & Deficiency Pathology

Vitamin	Chemical Name	Solubility Class	Deficiency Disease / Symptoms
Vitamin A	Retinol	Fat-Soluble	Night blindness (Nyctalopia), Xerophthalmia (corneal hardening)
Vitamin B_1	Thiamine	Water-Soluble	**Beriberi** (loss of appetite, muscular weakness, neuropathy)
Vitamin B_2	Riboflavin	Water-Soluble	Cheilosis (fissuring at corners of mouth), glossitis
Vitamin B_3	Niacin / Nicotinic acid	Water-Soluble	**Pellagra** (dermatitis, diarrhea, dementia)
Vitamin B_6	Pyridoxine	Water-Soluble	Convulsions, peripheral neuritis
Vitamin B_{12}	Cyanocobalamin	Water-Soluble	**Pernicious Anemia** (RBC deficiency; stored in liver!)
Vitamin C	Ascorbic Acid	Water-Soluble	**Scurvy** (bleeding gums, delayed wound healing; destroyed by heat)
Vitamin D	Ergocalciferol / Cholecalciferol	Fat-Soluble	**Rickets** in children, **Osteomalacia** in adults (bone softness)
Vitamin E	Tocopherols	Fat-Soluble	Muscular weakness, increased fragility of RBCs, loss of fertility
Vitamin K	Phylloquinone	Fat-Soluble	**Defective Blood Clotting** (prolonged prothrombin time)

Endocrine Hormones: Chemical Classification

Hormones are intercellular chemical messengers secreted directly into the bloodstream by ductless endocrine glands:

- **Steroid Hormones (Lipid-Soluble, Cholesterol Derivatives):** Secreted by adrenal cortex and gonads: Testosterone, Estrogens (Estradiol), Progesterone, Cortisol, Aldosterone.
- **Peptide / Protein Hormones (Water-Soluble):**
  - **Insulin (51 amino acids):** Secreted by β-cells of islets of Langerhans in pancreas; promotes glucose uptake and conversion to glycogen; deficiency causes *Diabetes mellitus*.
  - **Glucagon:** Secreted by α-cells of pancreas; stimulates glycogenolysis to elevate blood glucose.
  - **Oxytocin & Vasopressin (ADH):** Nonapeptides released by posterior pituitary gland.
- **Amine Hormones (Amino Acid Derivatives):**
  - **Epinephrine (Adrenaline) & Norepinephrine:** Derived from tyrosine; secreted by adrenal medulla ("fight or flight" hormones).
  - **Thyroxine (T_4):** Iodinated derivative of tyrosine produced by thyroid gland; regulates basal metabolic rate (BMR); deficiency causes goiter and hypothyroidism.

10 Master Distinction Matrix & High-Yield Synthesis Archetypes

10.1 Exhaustive Chemical Distinction Matrix

High-Yield Laboratory Chemical Tests for Functional Groups

Test Name	Reagents	Target Species	Positive Observation / Chemistry
Lucas Test	Anhyd. $ZnCl_2$ + conc. HCl	1°, 2°, 3° Alcohols	3°: Immediate turbidity; 2°: in 5 min; 1°: on boiling only
Neutral $FeCl_3$	Neutral $FeCl_3$ solution	Phenols	Violet / purple coloration (ferric phenoxide complex)
Tollens' Test	Ammoniacal $AgNO_3$	All Aldehydes	Shining silver mirror (Ag ↓) on inner tube walls
Fehling's Test	Fehling A ($CuSO_4$) + Fehling B (alkaline tartrate)	Aliphatic Aldehydes	Red precipitate of cuprous oxide (Cu_2O ↓); aromatic aldehydes fail
Iodoform Test	I_2 + NaOH (or NaOI)	$CH_3C=O$ or $CH_3CH(OH)-$	Yellow crystalline precipitate of CHI_3 ↓ (m.p. 119°C)
2,4-DNP Test	2,4-Dinitrophenylhydrazine	Aldehydes & Ketones	Yellow, orange, or red crystalline precipitate
Carbylamine Test	$CHCl_3$ + alc. KOH, Δ	1° Amines	Extremely obnoxious, foul odor of carbylamine ($R-NC$)
Hinsberg Test	$C_6H_5SO_2Cl$ + KOH	1°, 2°, 3° Amines	1°: sol. in KOH; 2°: insol. in KOH; 3°: no reaction
Azo Dye Test	1. $NaNO_2$/HCl (0 − 5°C), 2. alkaline β-naphthol	1° Aromatic Amines	Brilliant scarlet / orange red azo dye
Bicarbonate Test	Sat. $NaHCO_3$ solution	Carboxylic Acids	Brisk effervescence of CO_2 ↑ gas (phenols fail, except picric acid)

10.2 Solved Master Synthesis Archetypes

Archetype 1: Multi-Step Synthesis with Stereochemical Retention

Problem: Convert Benzene into pure *p*-bromoaniline. Identify why direct bromination of aniline fails, and formulate the complete protected sequence. **Step-by-Step Solution:**

1. **Pitfall of Direct Bromination:** Aniline contains a powerfully activating $-NH_2$ group ($+M$). Bromination with Br_2/H_2O yields instantaneous white precipitate of 2,4,6-tribromoaniline; monosubstitution cannot be controlled.
2. **Step 1 (Nitration):** Benzene $\xrightarrow[323\text{–}333\ K]{\text{conc. } HNO_3+H_2SO_4}$ $C_6H_5NO_2$ (Nitrobenzene).
3. **Step 2 (Reduction):** $C_6H_5NO_2 \xrightarrow{Sn/HCl \text{ or } Fe/HCl}$ $C_6H_5NH_2$ (Aniline).
4. **Step 3 (Protection via Acetylation):** $C_6H_5NH_2 + (CH_3CO)_2O \xrightarrow{\text{pyridine}}$ $C_6H_5NHCOCH_3$ (Acetanilide). The lone pair on nitrogen is delocalized into the acetyl carbonyl, moderating ring activation.
5. **Step 4 (Regioselective Bromination):** $C_6H_5NHCOCH_3 + Br_2 \xrightarrow{CH_3COOH}$ *p*-bromoacetanilide (exclusive para product due to steric hindrance at ortho positions).
6. **Step 5 (Deprotection):** *p*-bromoacetanilide $\xrightarrow{H_3O^+ \text{ or } OH^-, \Delta}$ **p-bromoaniline** + CH_3COOH.

Archetype 2: Carbon-Carbon Chain Lengthening & Functional Transformation

Problem: Formulate the synthetic conversion of Ethanol into Propan-1-amine and Propan-2-ol. **Step-by-Step Solution:**

1. **Conversion of Ethanol into Propan-1-amine (One Carbon Ascent):**

$$CH_3CH_2OH \xrightarrow{PCl_5 \text{ or } SOCl_2} CH_3CH_2Cl \quad \text{(Ethyl chloride)} \tag{97}$$

$$CH_3CH_2Cl \xrightarrow{\text{ethanolic KCN}} CH_3CH_2CN \quad \text{(Propanenitrile: nucleophilic substitution)} \tag{98}$$

$$CH_3CH_2CN \xrightarrow{LiAlH_4 \text{ or } Na/EtOH} CH_3CH_2CH_2NH_2 \quad \text{(Propan-1-amine: complete reduction)} \tag{99}$$

2. **Conversion of Ethanol into Propan-2-ol (Grignard Pathway):**

$$CH_3CH_2OH \xrightarrow{PCC/CH_2Cl_2} CH_3CHO \quad \text{(Acetaldehyde: selective oxidation)} \tag{100}$$

$$CH_3CHO + CH_3MgBr \xrightarrow{\text{dry ether}} [CH_3 - CH(OMgBr) - CH_3] \tag{101}$$

$$[CH_3 - CH(OMgBr) - CH_3] \xrightarrow{H_3O^+} CH_3 - CH(OH) - CH_3 \quad \text{(Propan-2-ol)} \tag{102}$$

Master Summary: Key Takeaways for NEET UG 2027 Organic Chemistry

- **Always inspect carbocation intermediates** for possible 1,2-hydride, 1,2-methyl, or ring-expansion rearrangements before forming products in $S_N1, E1$, and electrophilic addition reactions.
- **In Williamson ether synthesis**, always employ the alkoxide of the bulky tertiary component and a primary alkyl halide to avoid elimination.
- **In biomolecules**, remember that sucrose is non-reducing and lacks mutarotation; all monosaccharides, maltose, and lactose are reducing.
- **In DNA double helix**, $A = T$ (2 H-bonds) and $G \equiv C$ (3 H-bonds); Chargaff's rules hold strictly for double-stranded DNA.
- **In aromatic synthesis**, protect aniline by acetylation with acetic anhydride before nitration to avoid the anomalous 47% meta-product.